Capítulo 13 – Elementos de Máquinas

OBJECTIVOS

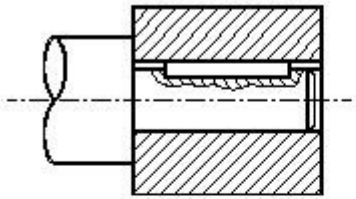
- Compreender a representação de elementos normalizados;
- Representar, cotar e referenciar elementos de máquinas;
- Distinguir e compreender formas de ligação;
- Distinguir os elementos normalizados na representação de conjunto num desenho.

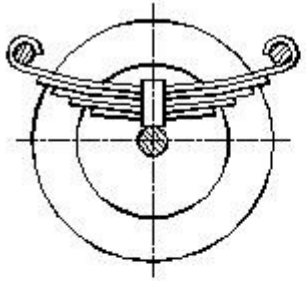
Silva, Ribeiro, Dias, Sousa

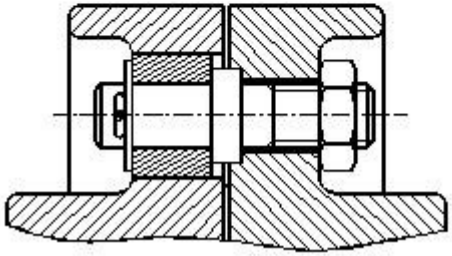
ELEMENTOS DE MÁQUINAS

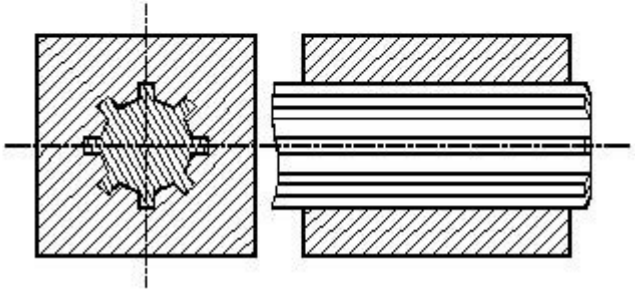
PROCESSOS DE LIGAÇÃO

- Permanentes
  - -directas
  - -indirectas
- desmontáveis





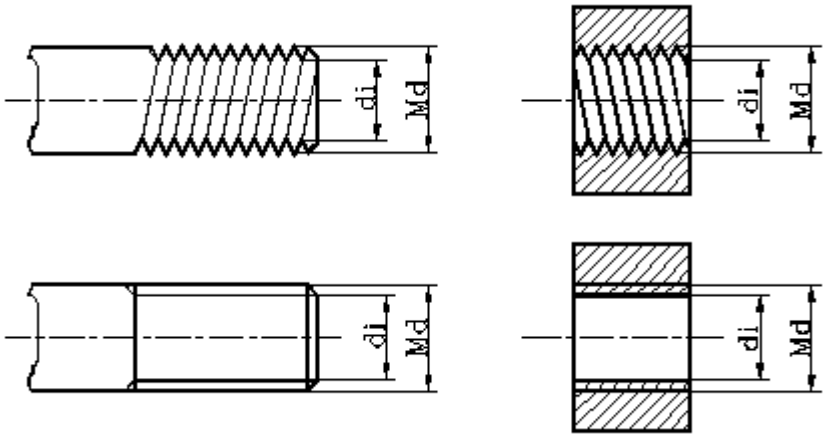


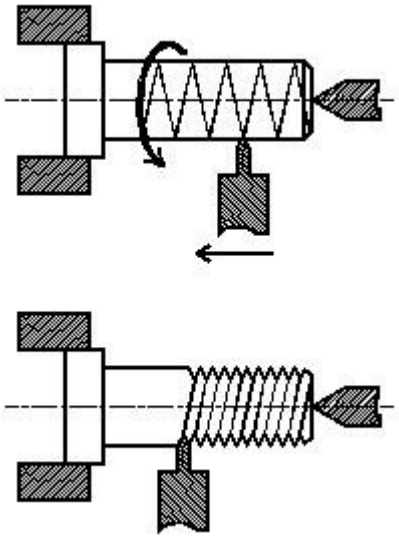


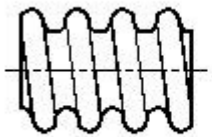
ELEMENTOS DE MÁQUINAS

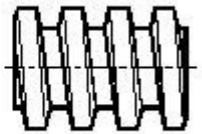
ROSCAS

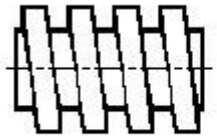
- Geometria
- Profundidade
- Passo

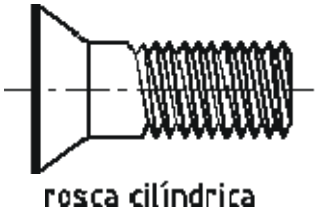


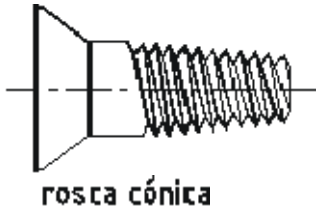






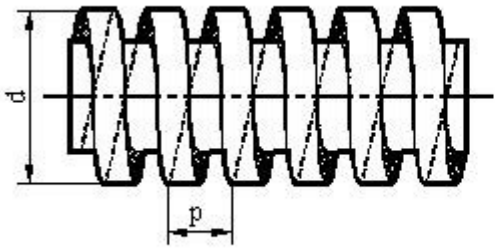




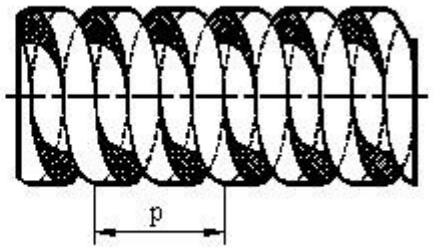


ELEMENTOS DE MÁQUINAS

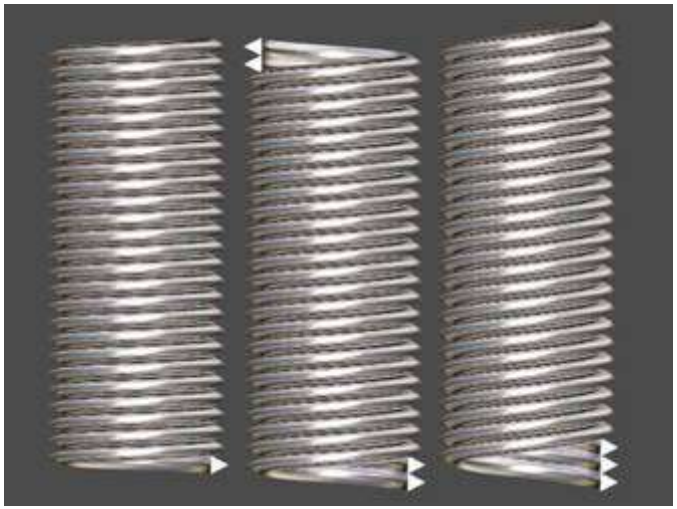
- Rosca simples
- Rosca múltipla



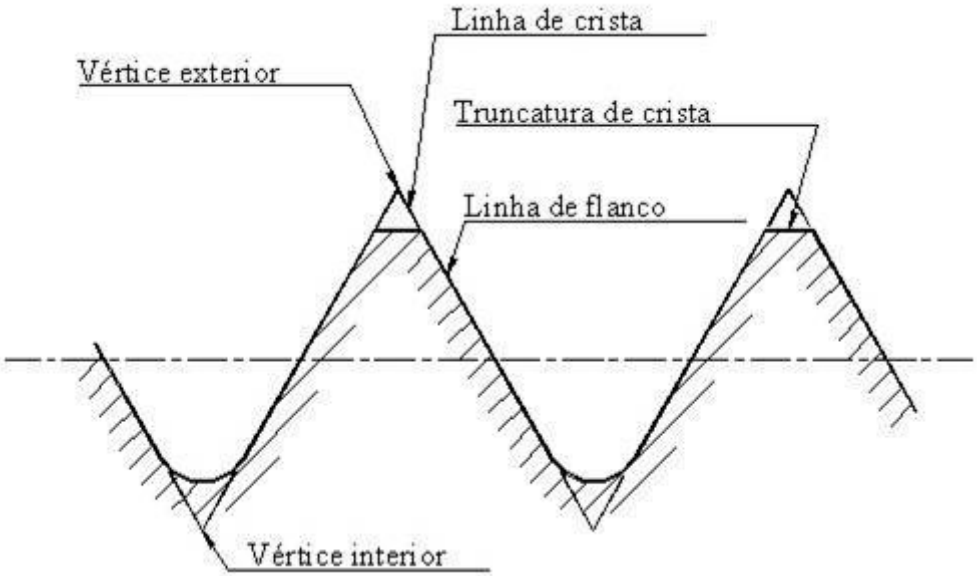
Rosca simples



Rosca dupla (2 entradas)



- Perfil triangular
  - -Flanco
  - -Crista
  - -Cava



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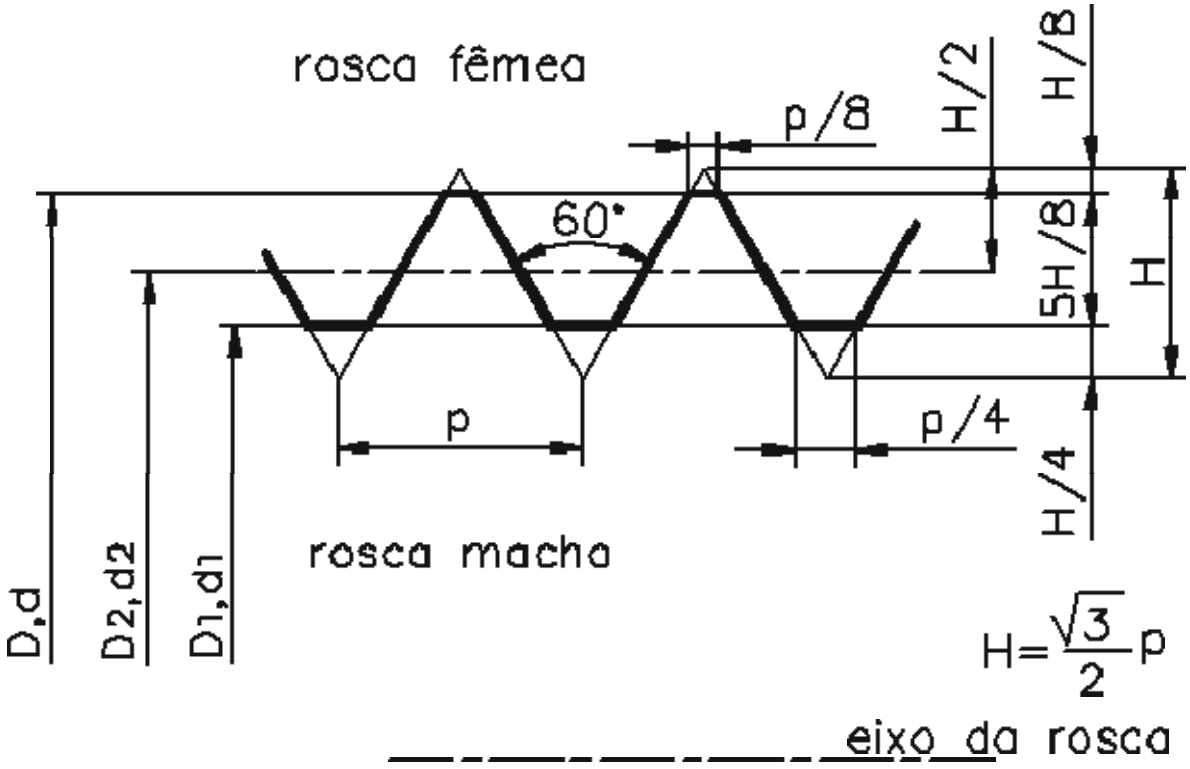
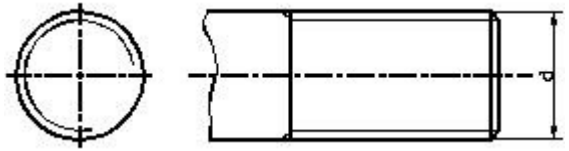
ELEMENTOS DE MÁQUINAS

Elementos dimensionais de uma rosca triangular

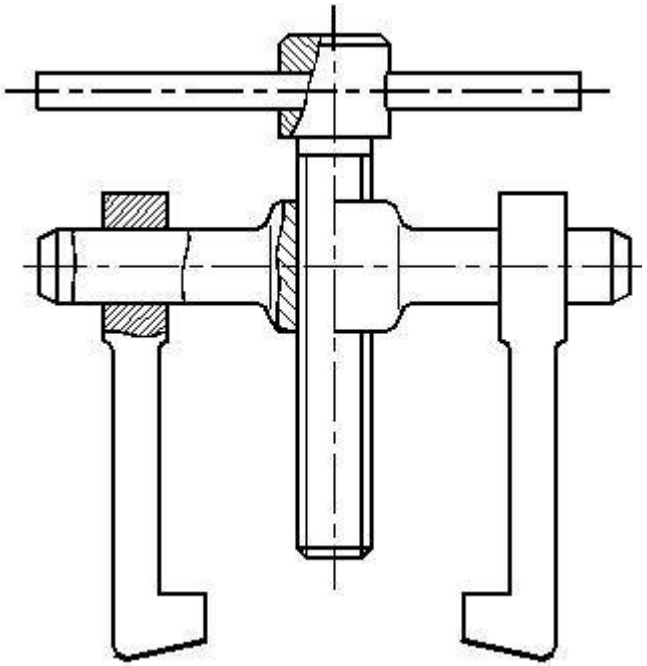


ELEMENTOS DE MÁQUINAS

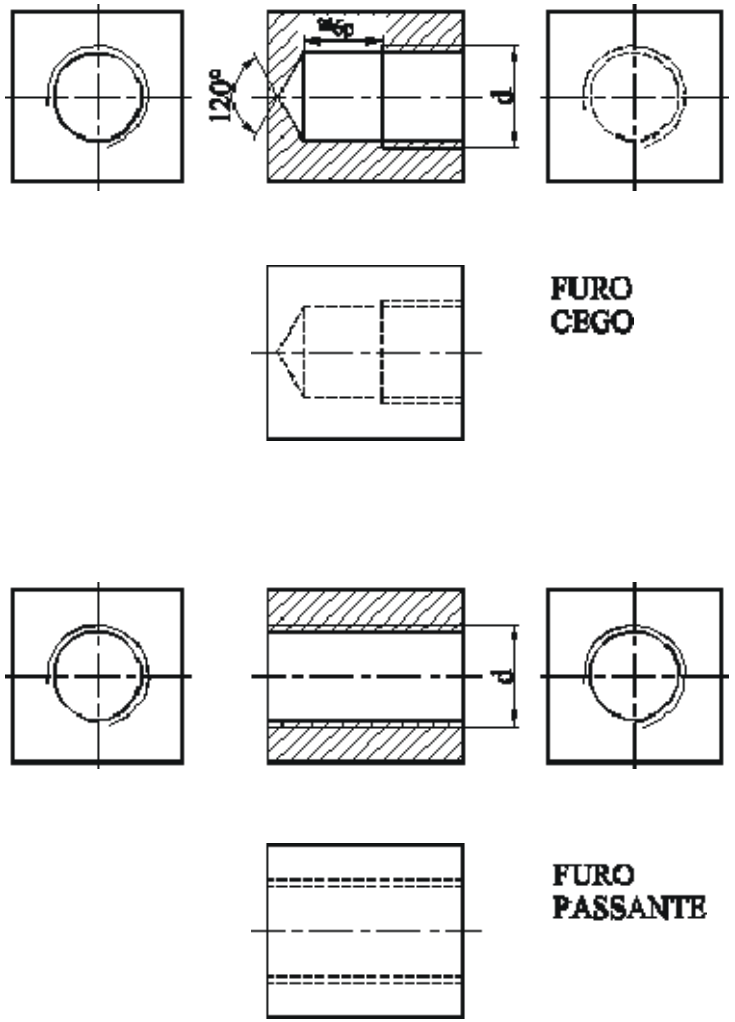
Representação de roscas



Conjunto de peças roscadas



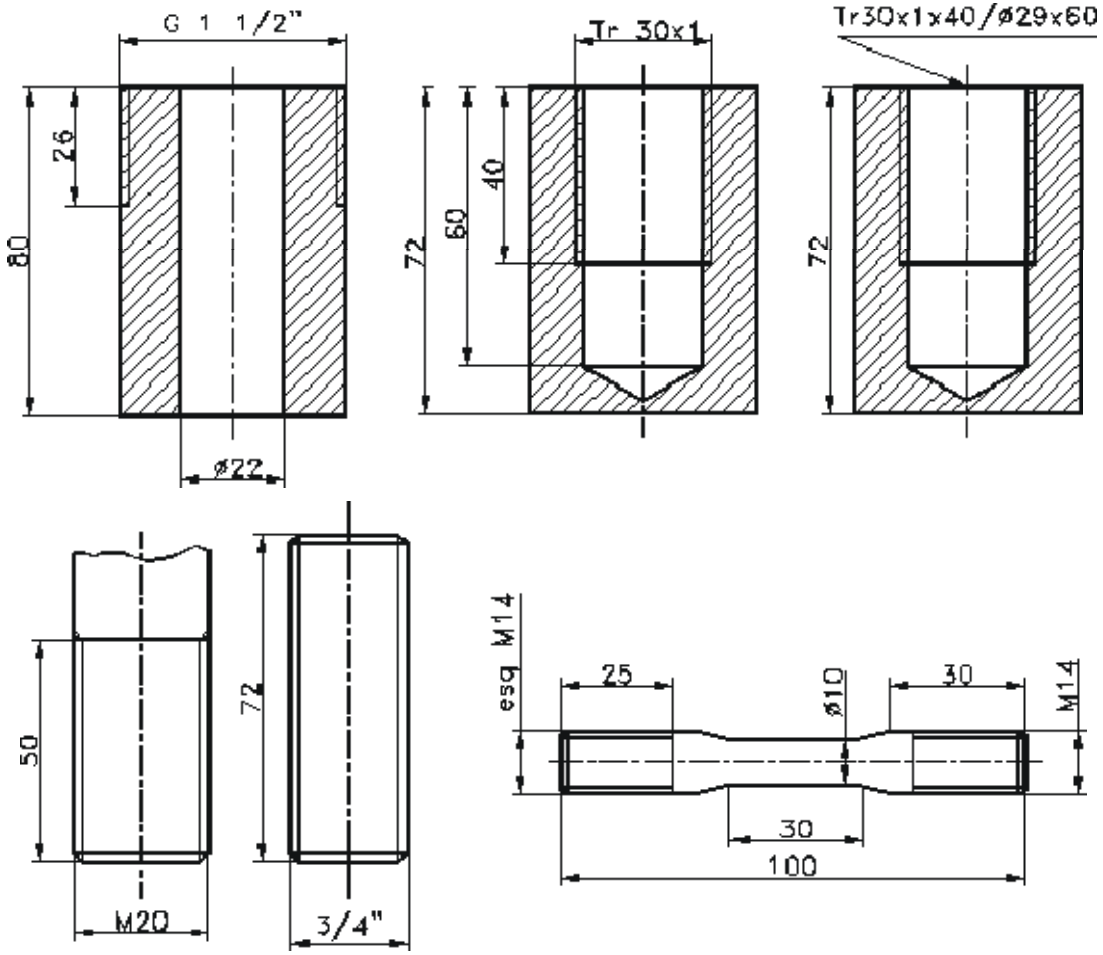
Furos



ELEMENTOS DE MÁQUINAS

Cotagem de Roscas

- **ISO** (métrica) (M)
- *Whitworth*
- Gás (G)
- Rectangular (R)
- Trapezoidal (Tr)
- Dente de serra (S)
- Redonda (Rd)



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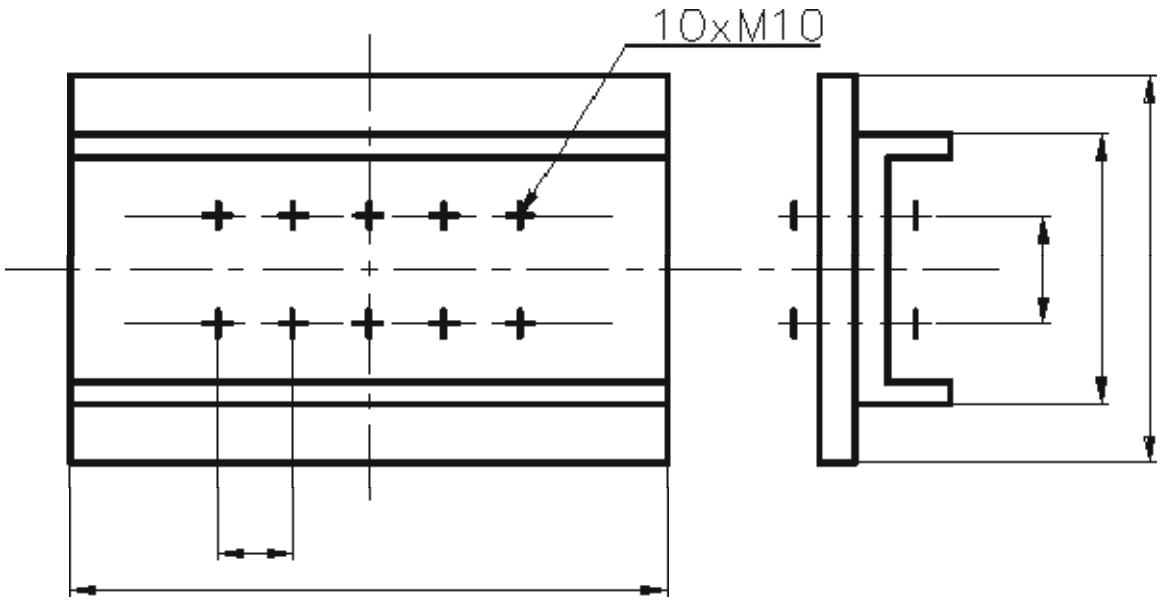
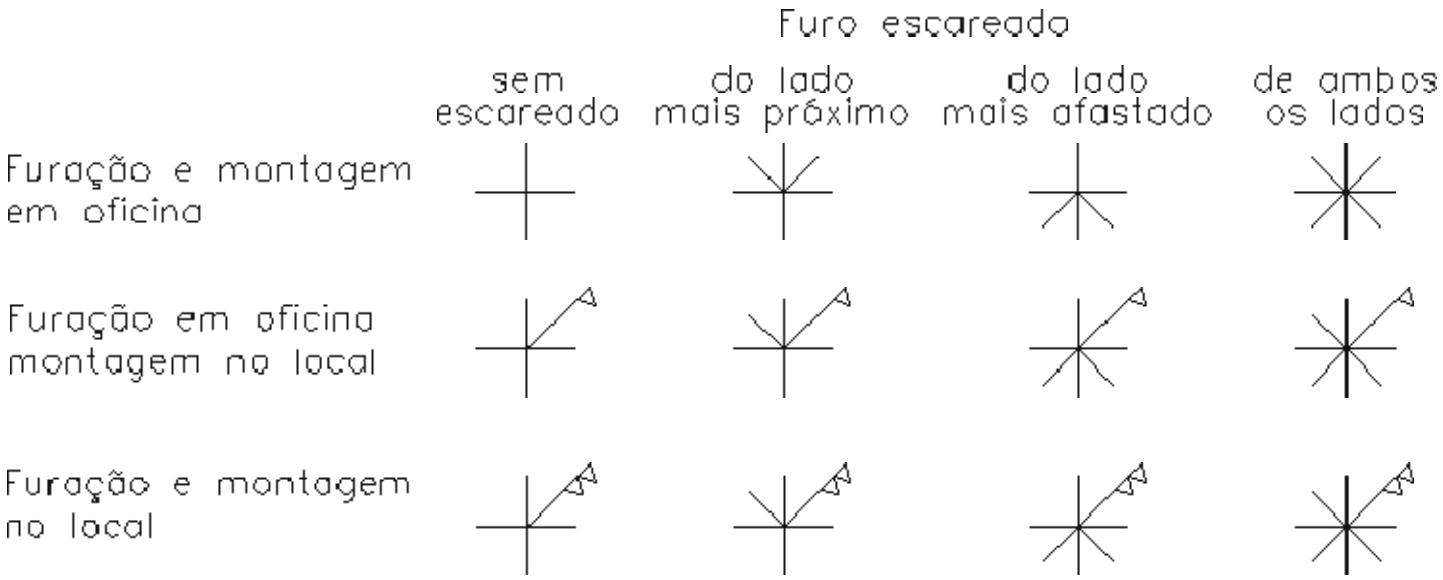
ELEMENTOS DE MÁQUINAS

Representação de furações

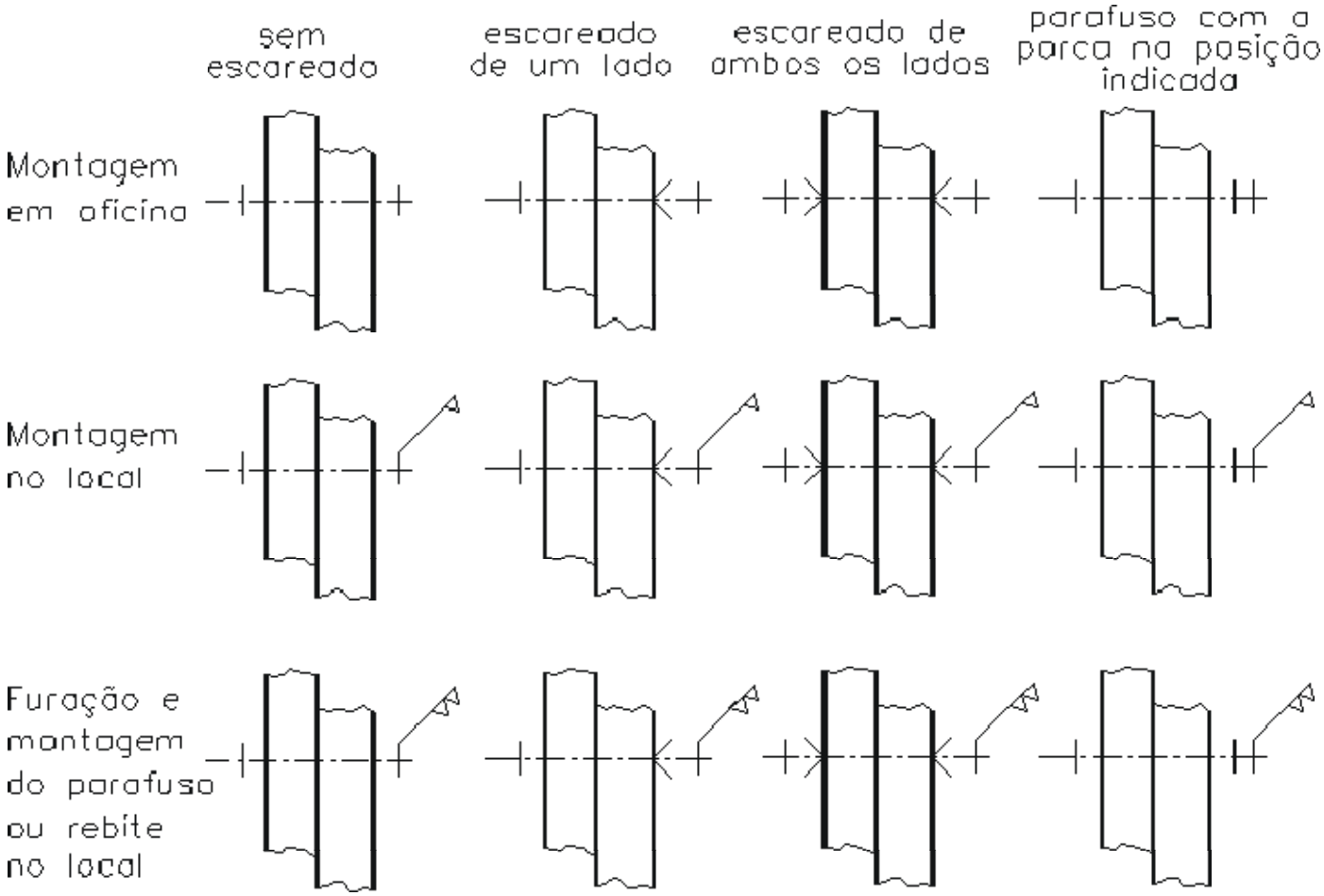


Representação simbólica de furos



ELEMENTOS DE MÁQUINAS

Representação simbólica de furos e ligações

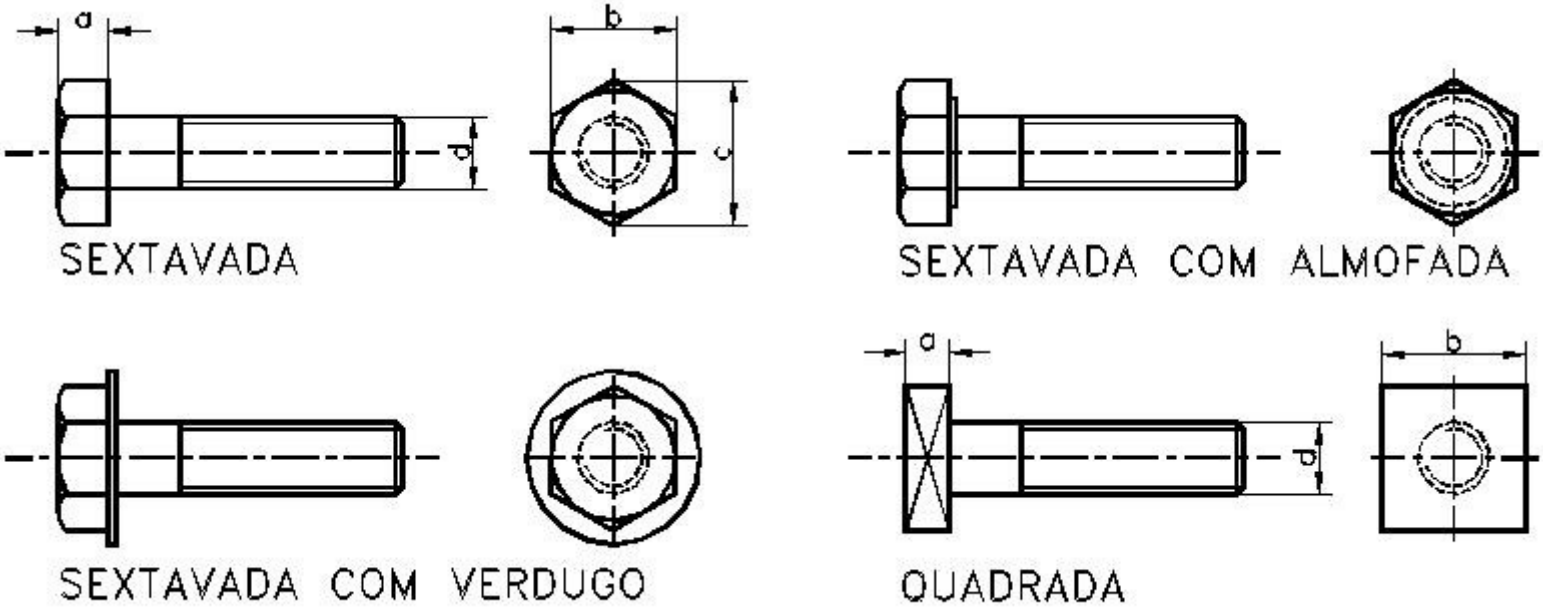


Parafusos

- Forma da Cabeça (ISO1891)
  Prismática,
  Fenda,
  Caixa,
  outros
- Haste ou Espiga (ISO4753)

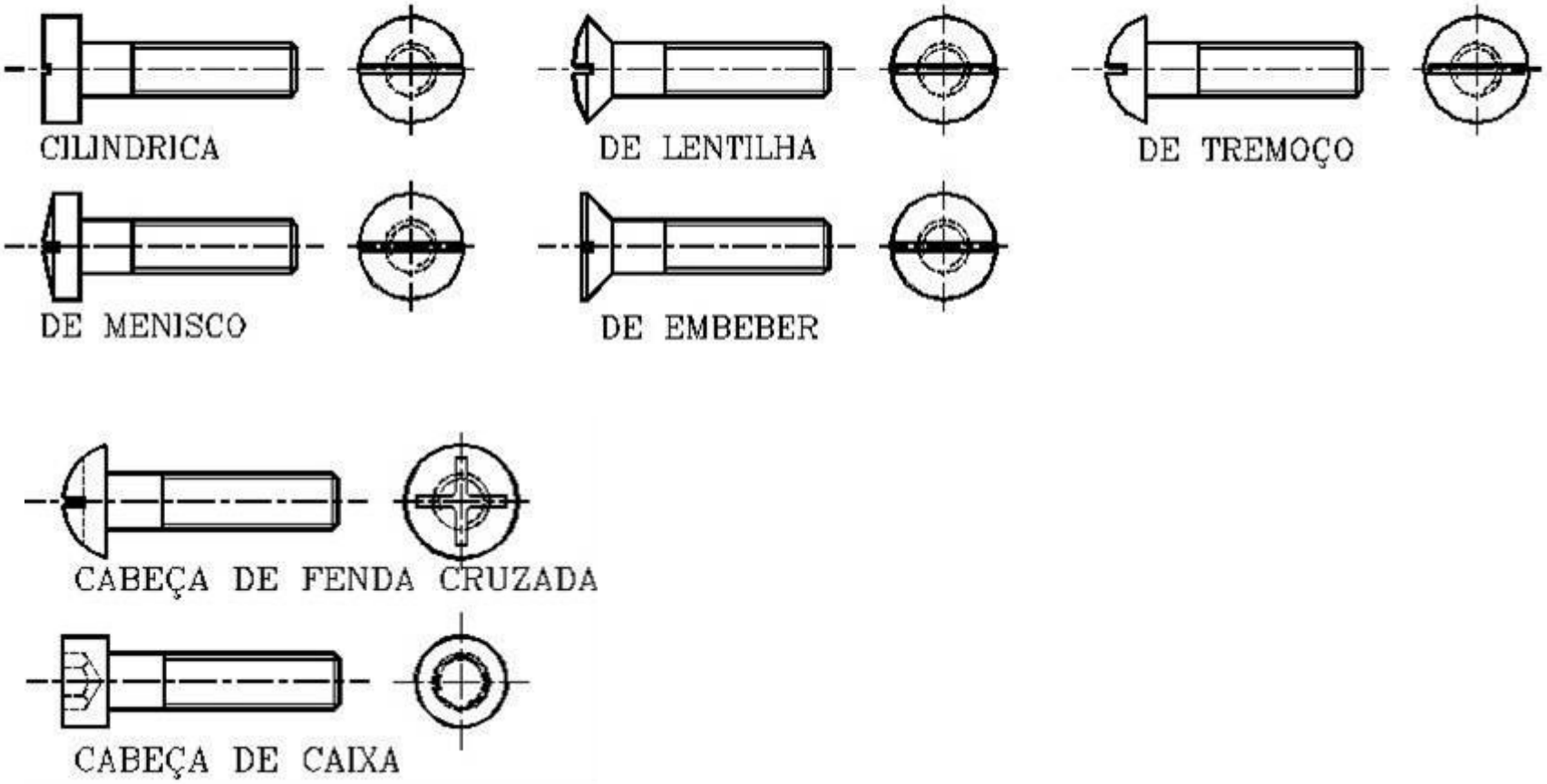
Cabeças de Parafusos

Prismática



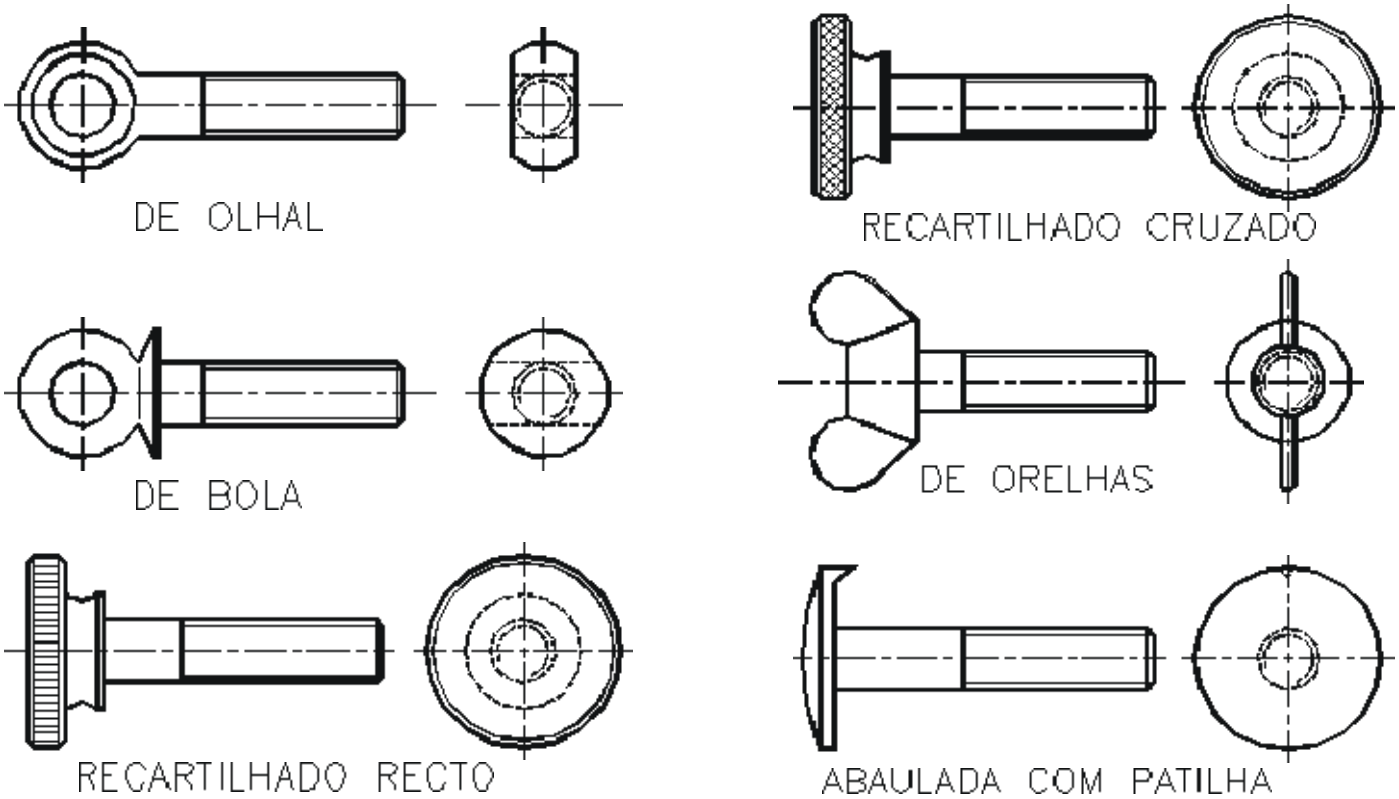
ELEMENTOS DE MÁQUINAS

Fenda



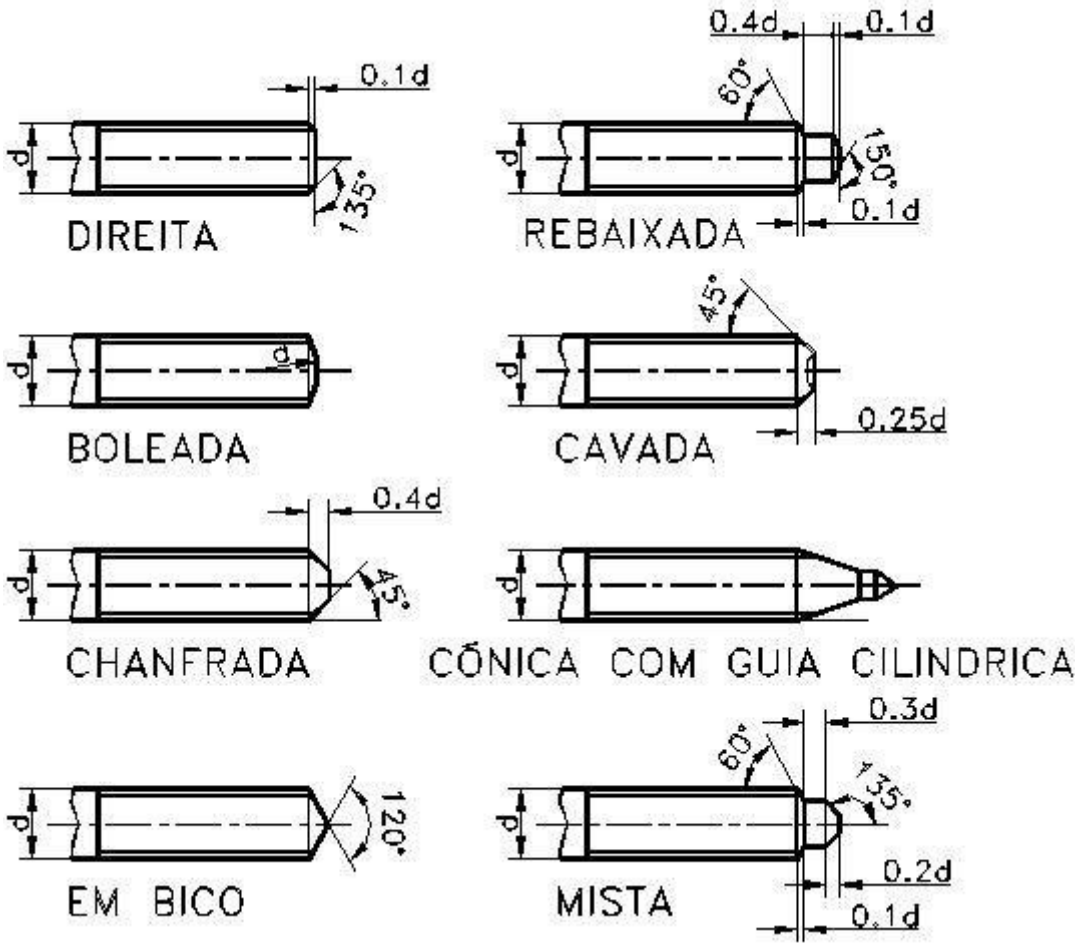
ELEMENTOS DE MÁQUINAS

outros tipos

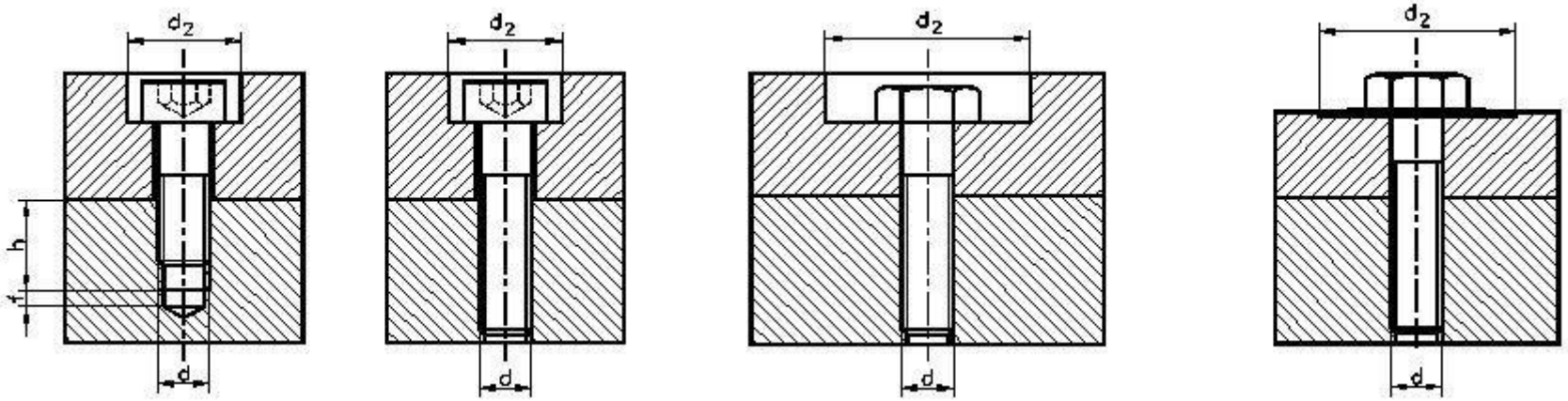


ELEMENTOS DE MÁQUINAS

Espiga

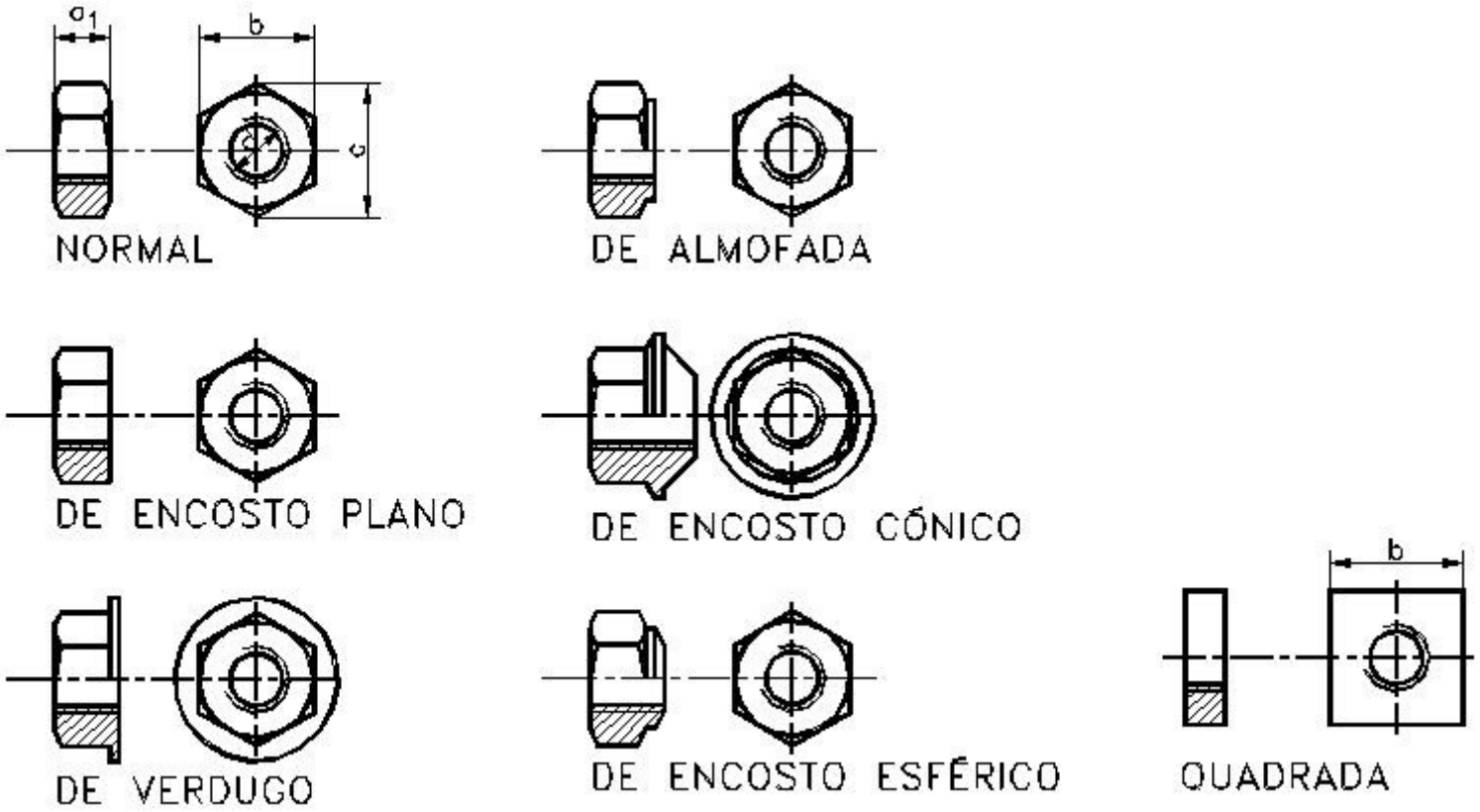


Caixa



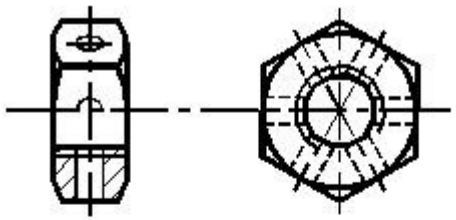
Porcas

Prismáticas para aperto com chave

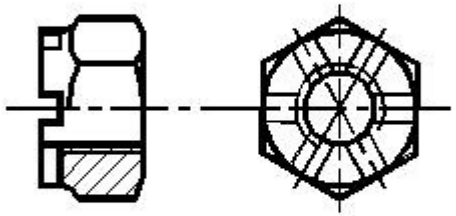


ELEMENTOS DE MÁQUINAS

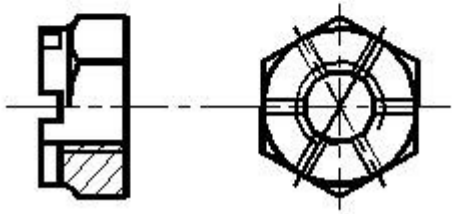
Para imobilização



COM FUROS PARA TROÇO

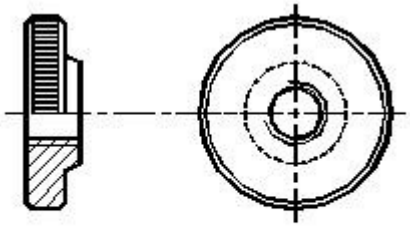


DE CASTELO

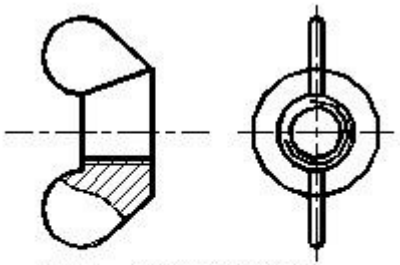


SERRADA

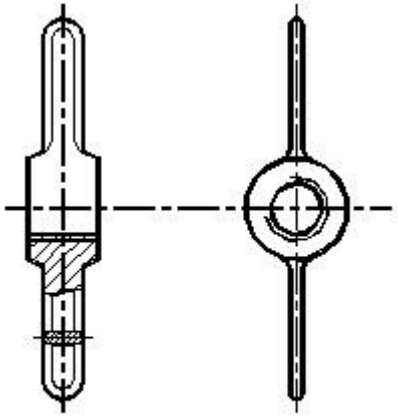
Para apertar à mão



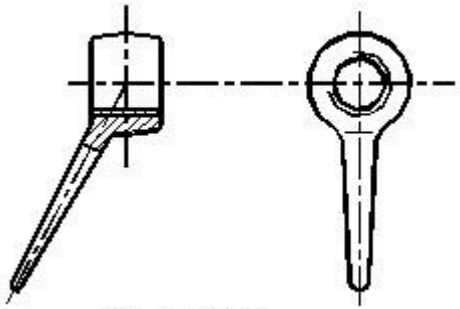
RECARTILHADA



DE ORELHAS



DE MANÍPULO



DE RABO

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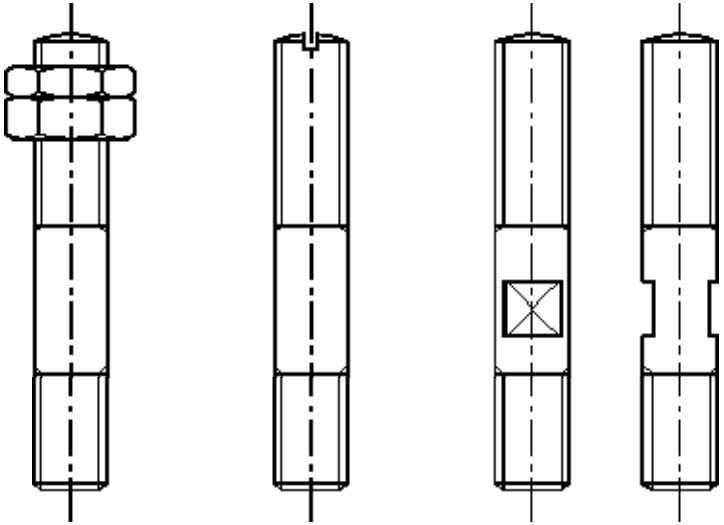
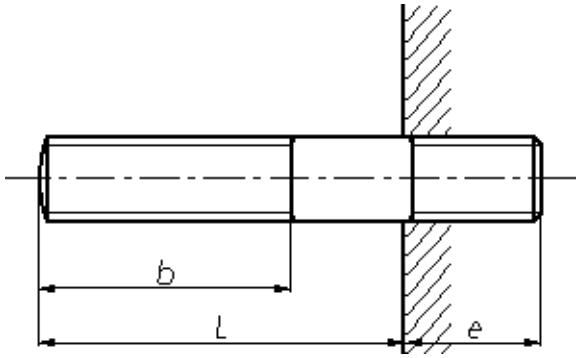
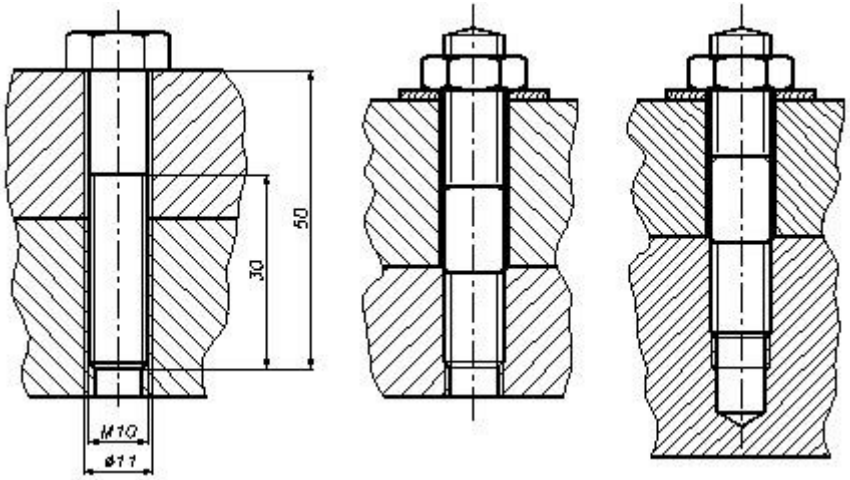
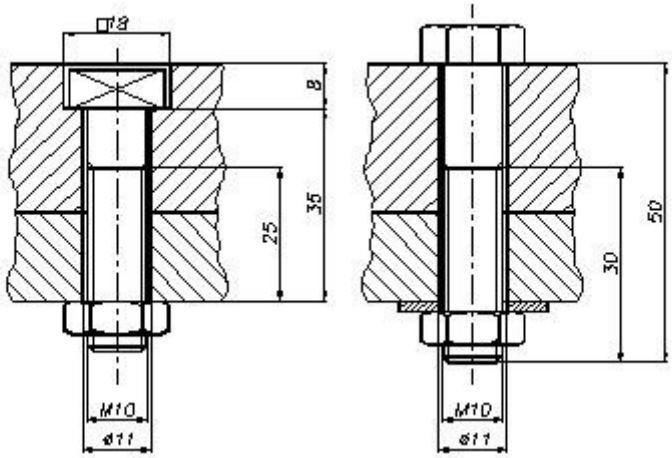
ELEMENTOS DE MÁQUINAS

Pernos - haste cilíndrica roscada
em ambas as extremidades

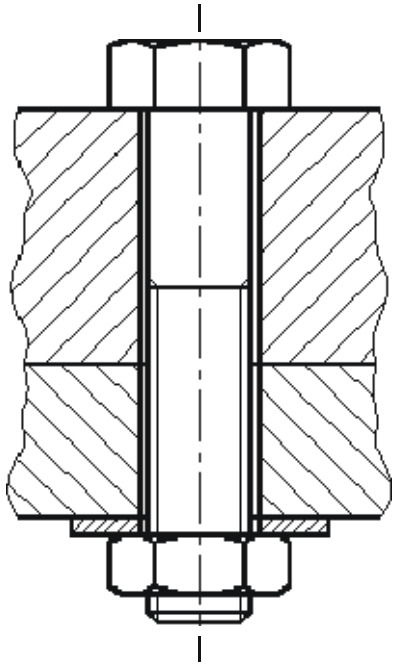


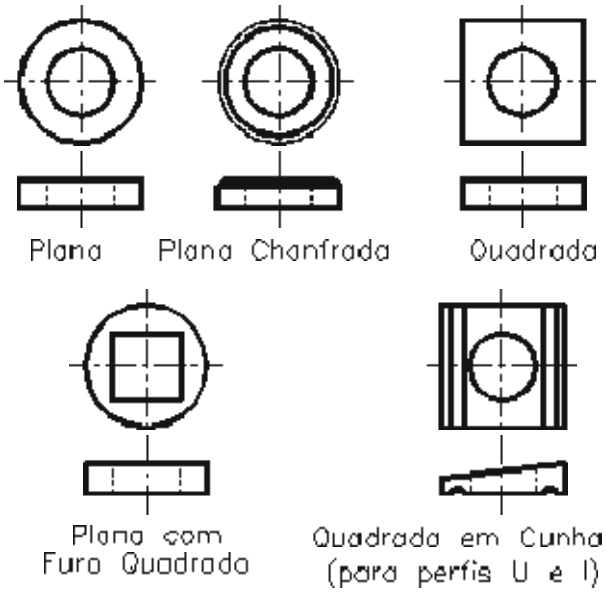


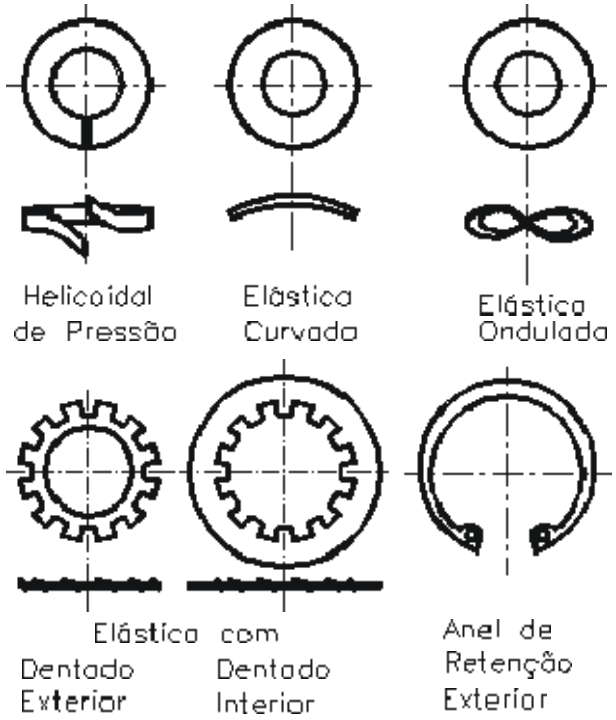


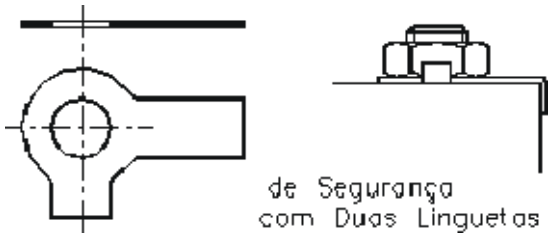


Anilhas

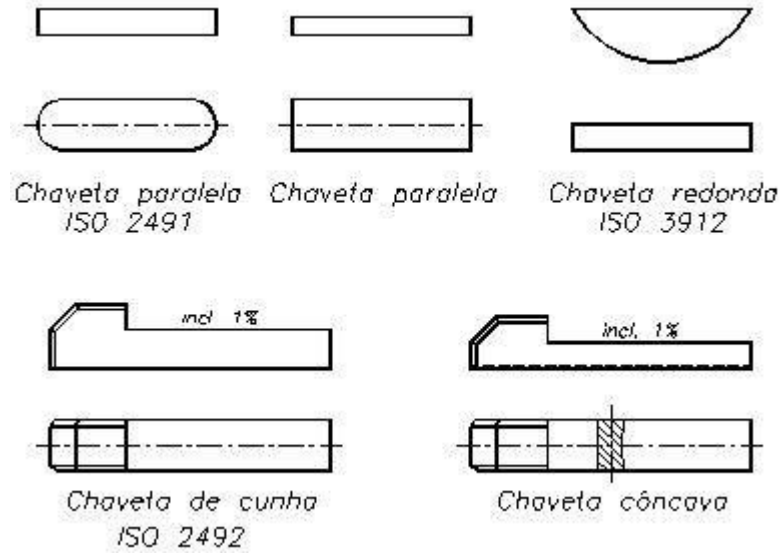


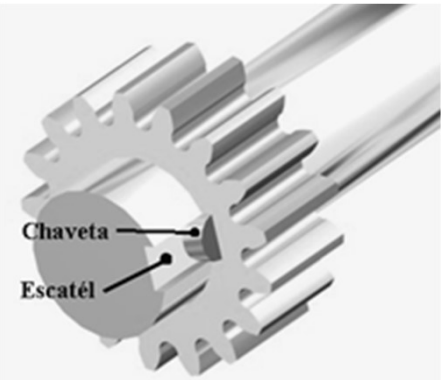

Plana
Plana Chanfrada
Quadrada
Plana com Furo Quadrado
Quadrada em Cunha (para perfis U e I)

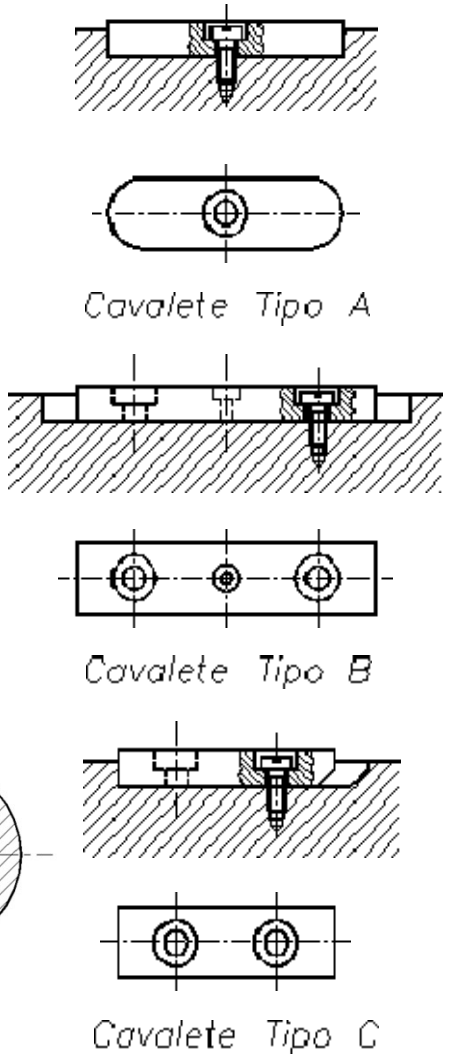

Helicoidal de Pressão
Elástica Curvada
Elástico Ondulado
Elástica com Dentado Exterior
Dentado Interior
Anel de Retenção Exterior

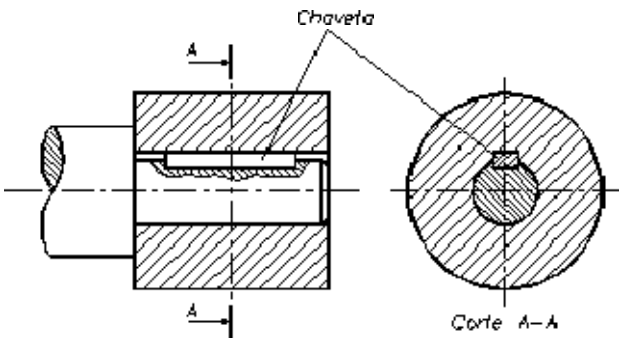

de Segurança com Duas Linguetas

Chavetas e Cavaletes

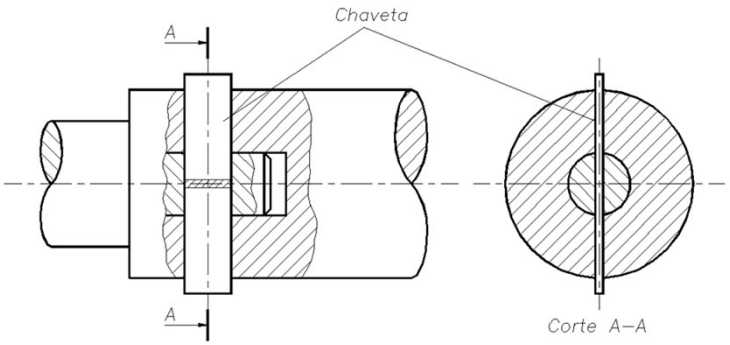






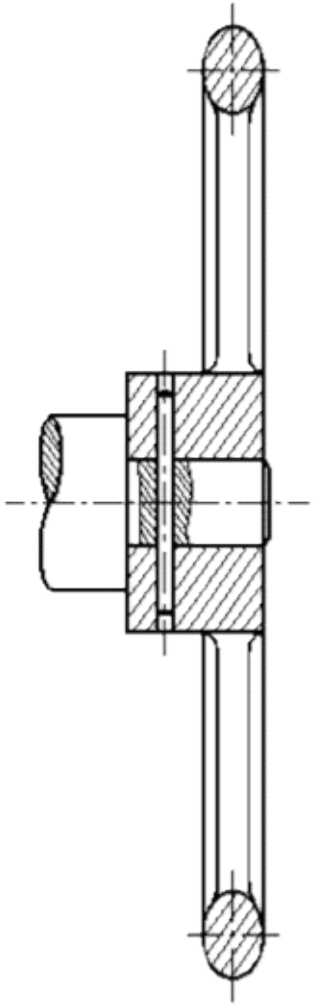


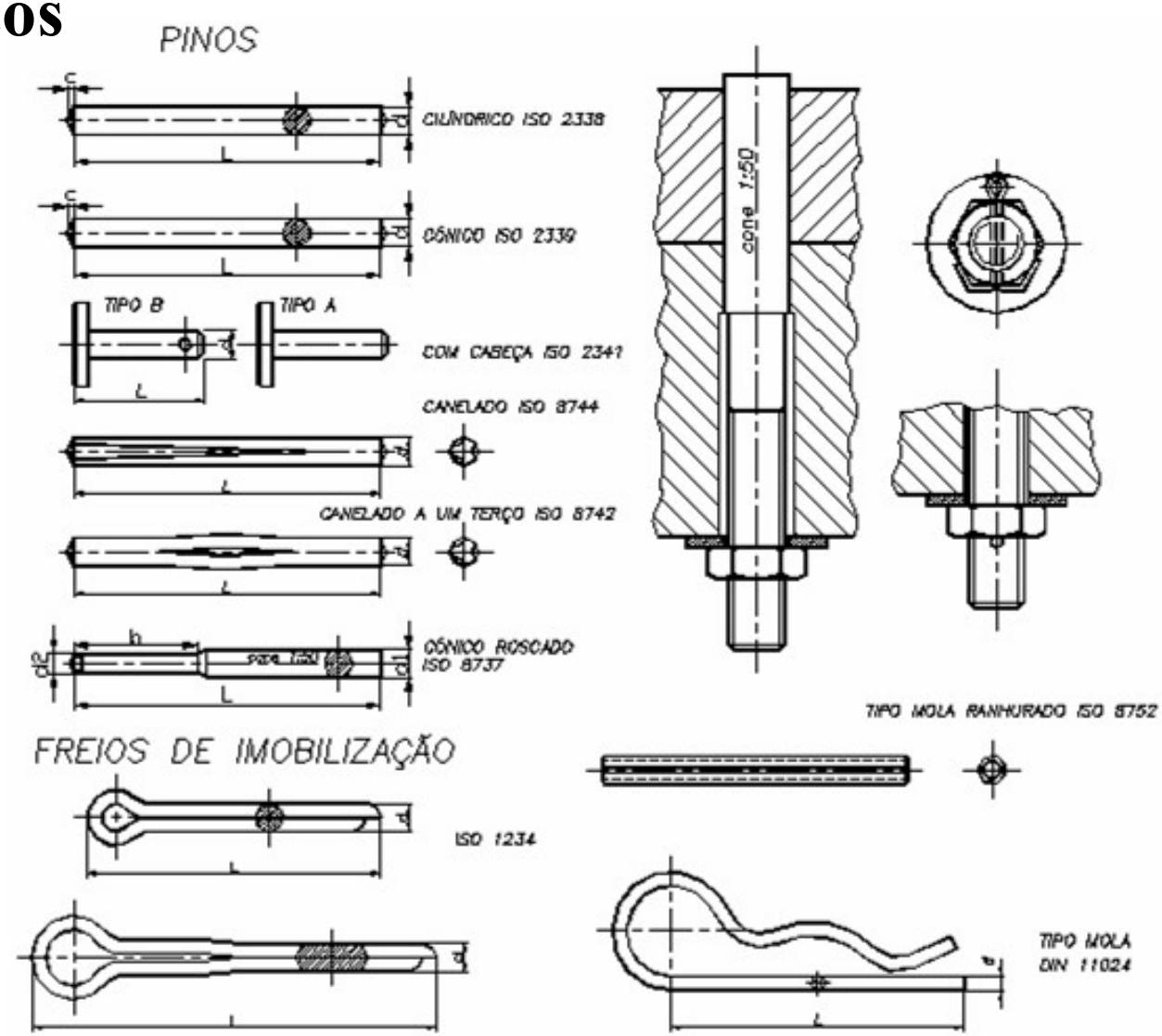
longitudinais



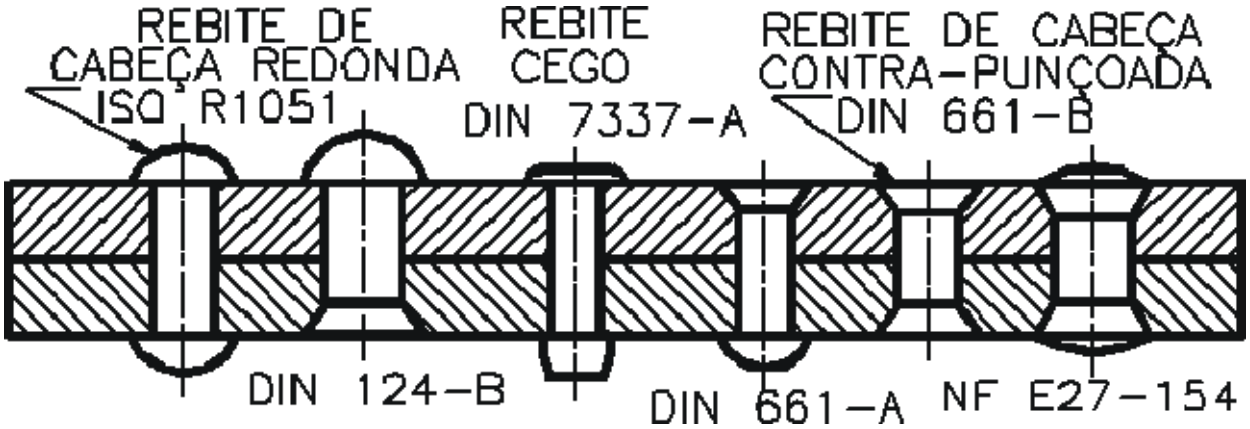
transversais

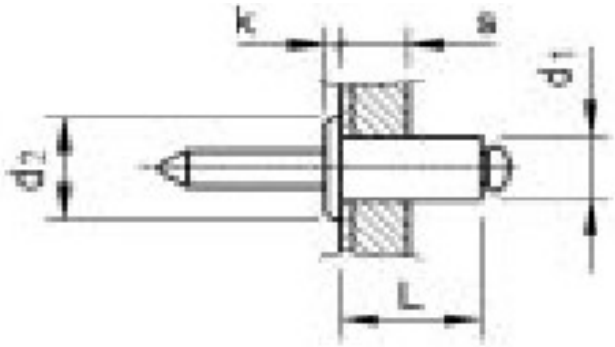
Cavilhas e Troços





Rebites

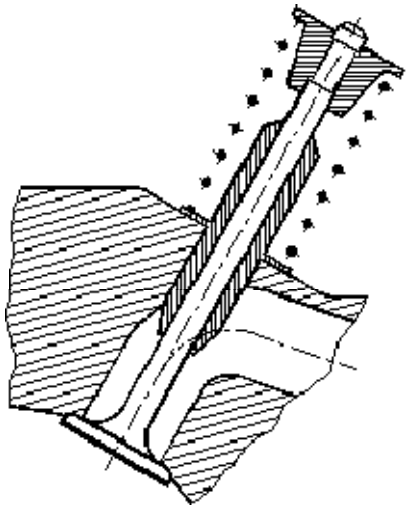

REBITE DE CABEÇA REDONDA ISO R1051
REBITE CEGO DIN 7337-A
REBITE DE CABEÇA CONTRA-PUNÇOADA DIN 661-B
DIN 124-B
DIN 661-A
NF E27-154

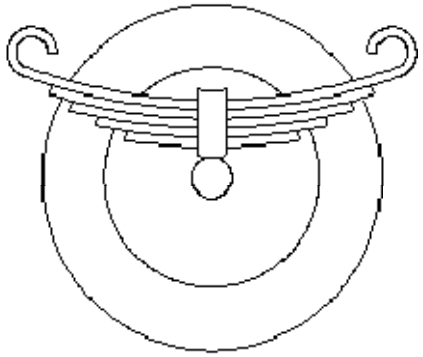

k
s
d1
d2
L

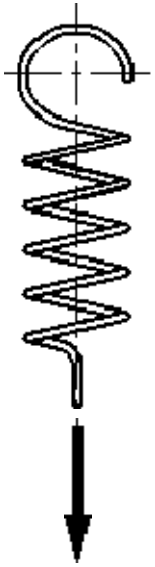
ELEMENTOS DE MÁQUINAS

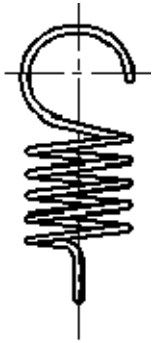
Molas

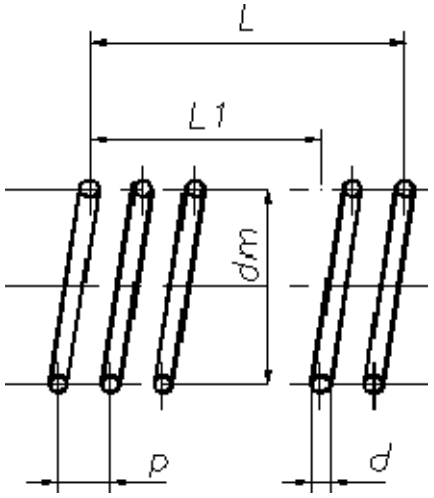
- Helicoidal
- Voluta
- Anilhas Elásticas
- Espiral
- Folha ou Lâminas



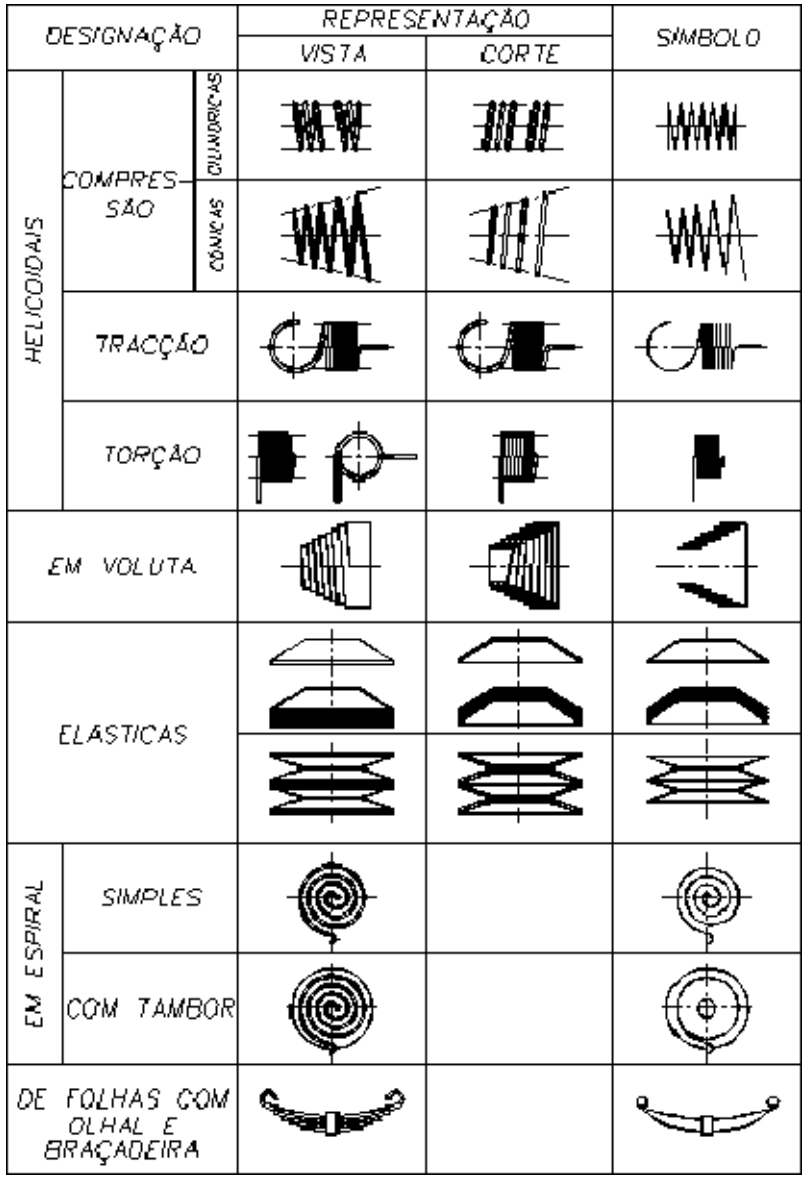








ELEMENTOS DE MÁQUINAS

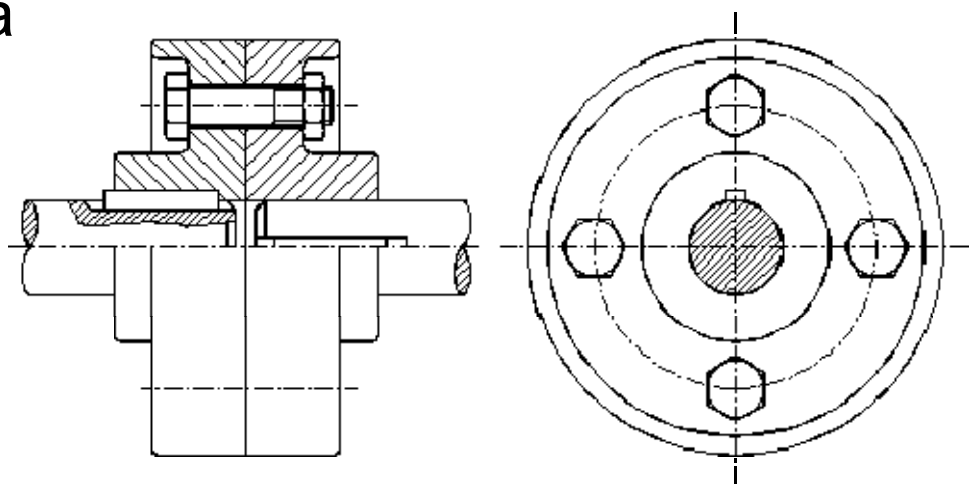


Representação de Molas (ISO 2162)

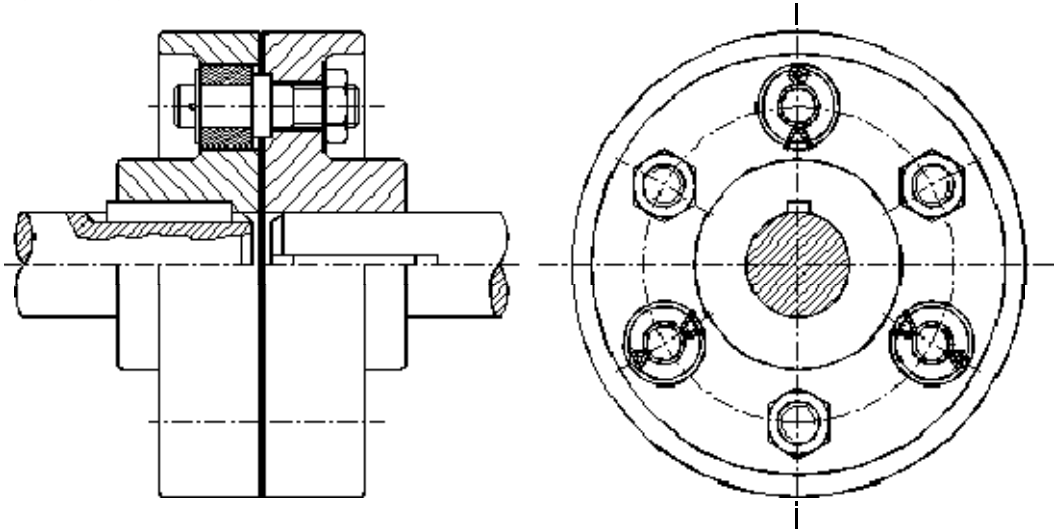
ELEMENTOS DE MÁQUINAS

Uniões de Veios

Fixa

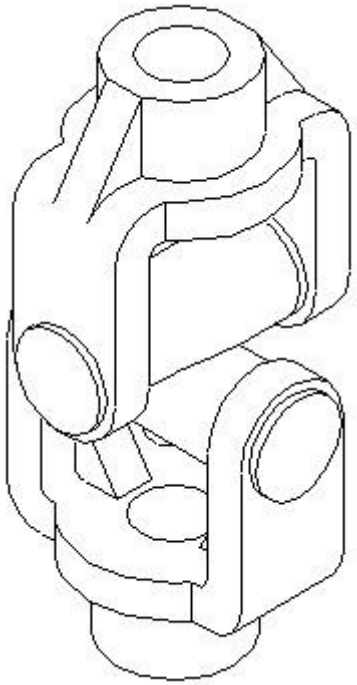


Elástica

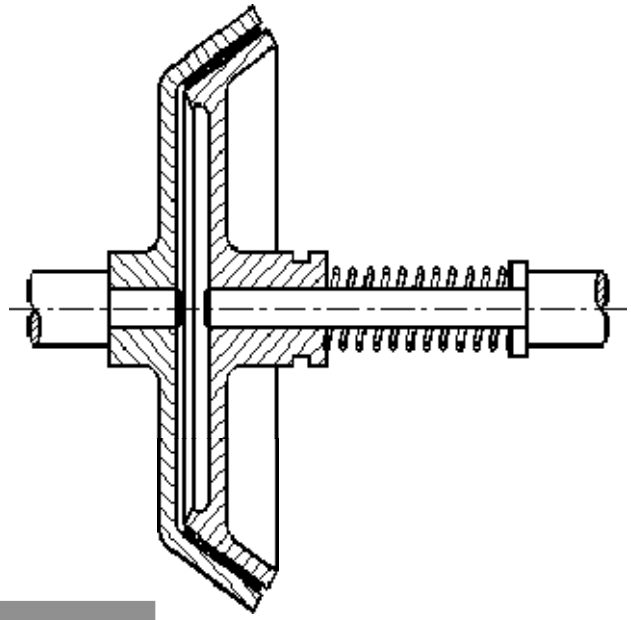


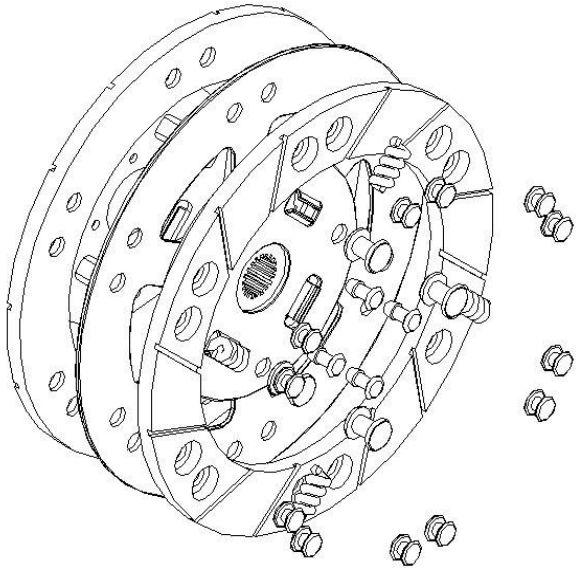
ELEMENTOS DE MÁQUINAS

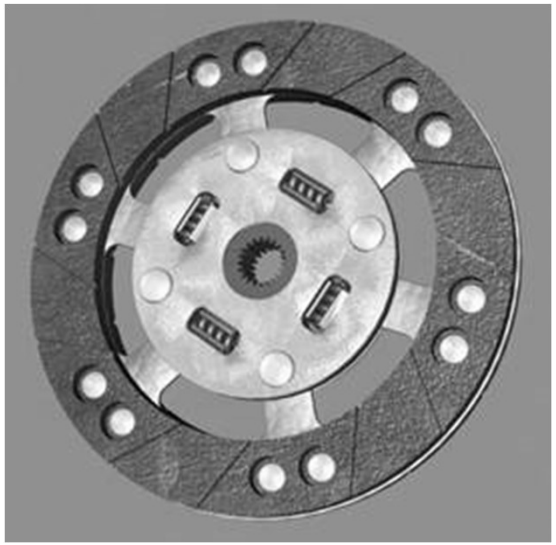
Cardan



Embraiagens

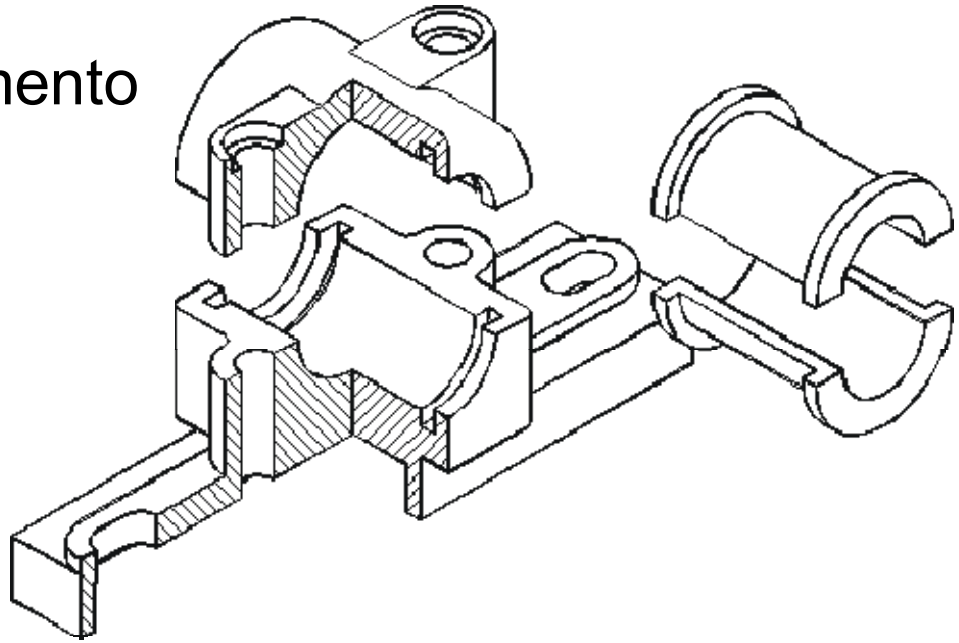


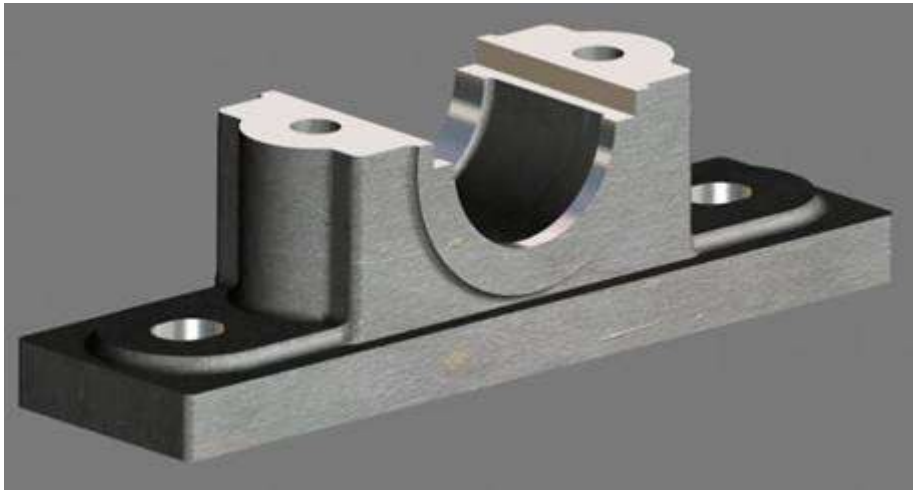




ELEMENTOS DE MÁQUINAS

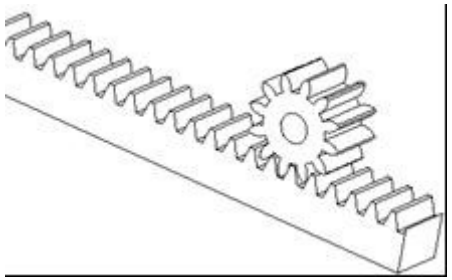
Chumaceira de escorregamento

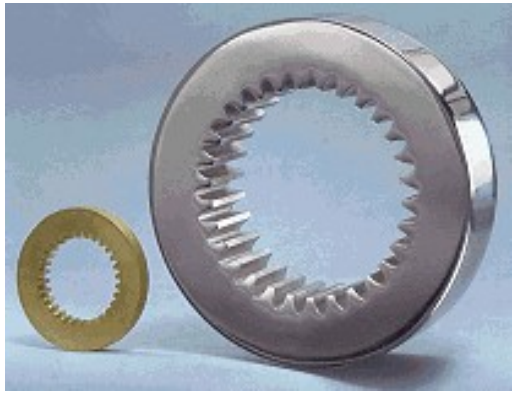




ELEMENTOS DE MÁQUINAS

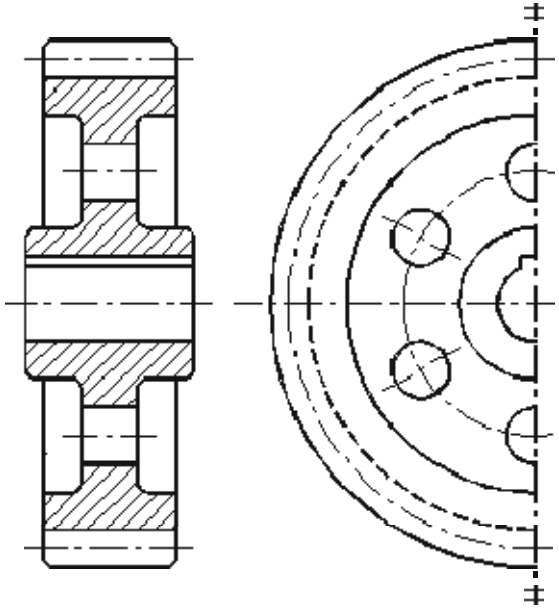
Engrenagens

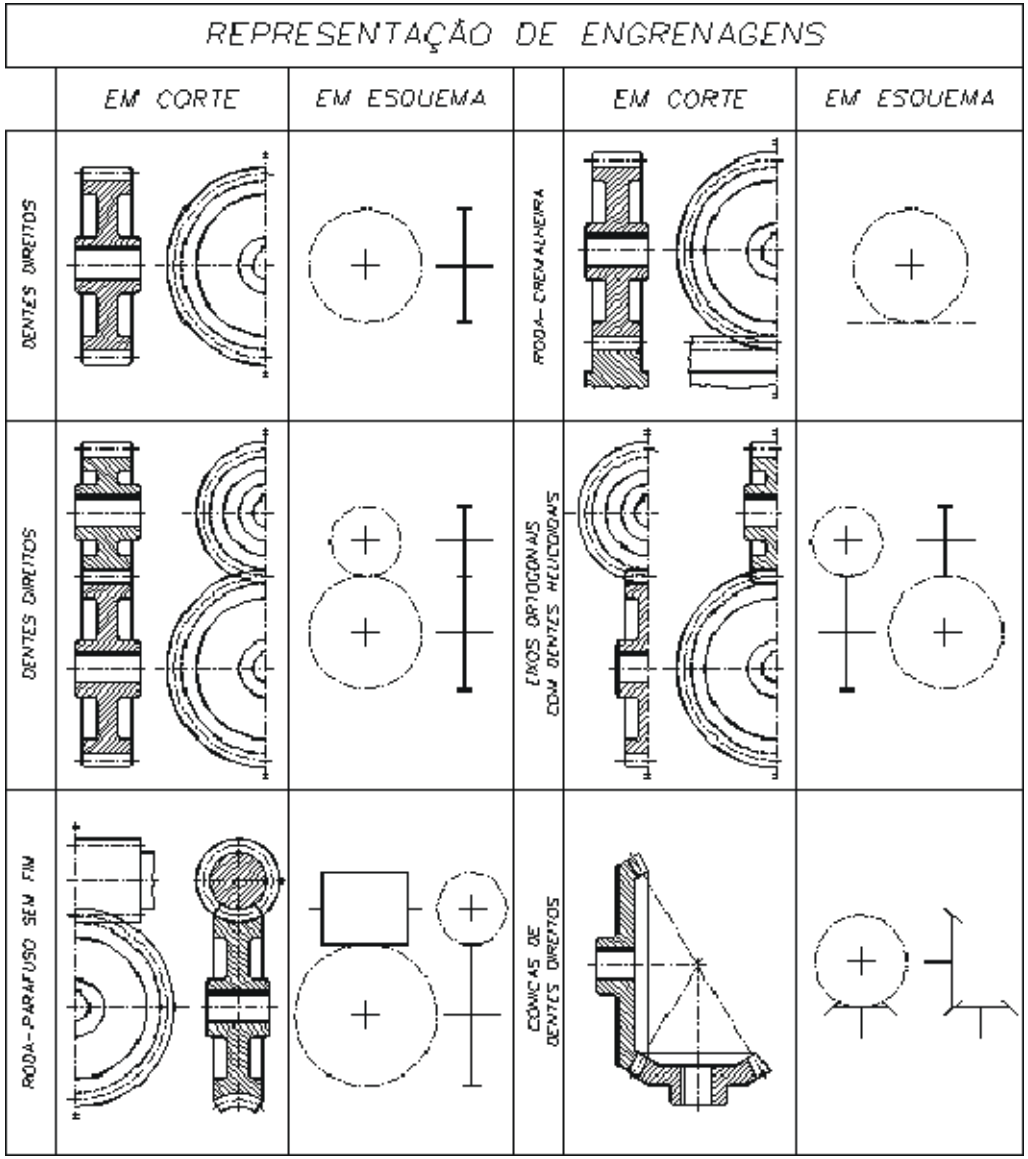




ELEMENTOS DE MÁQUINAS

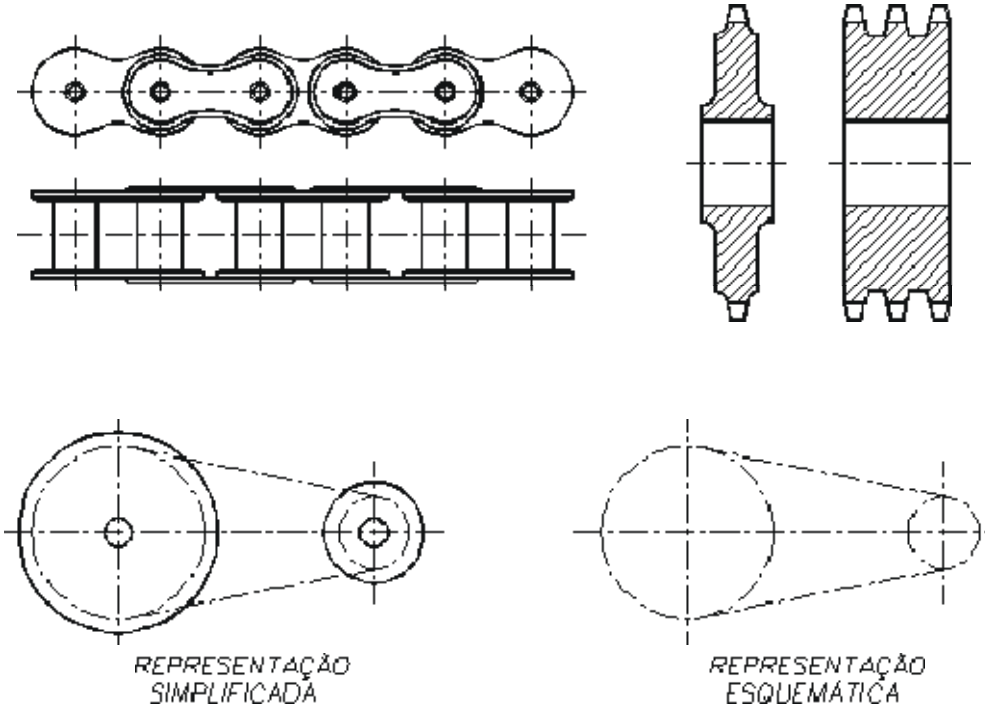
Representação de Engrenagens



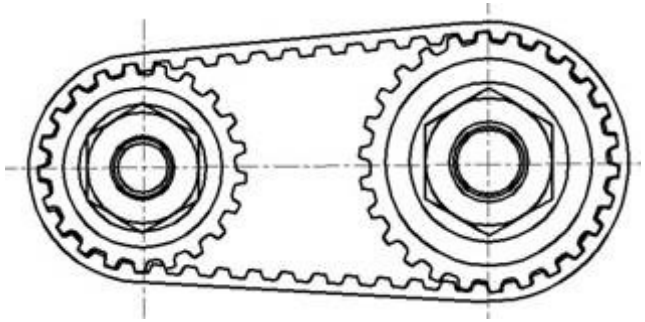


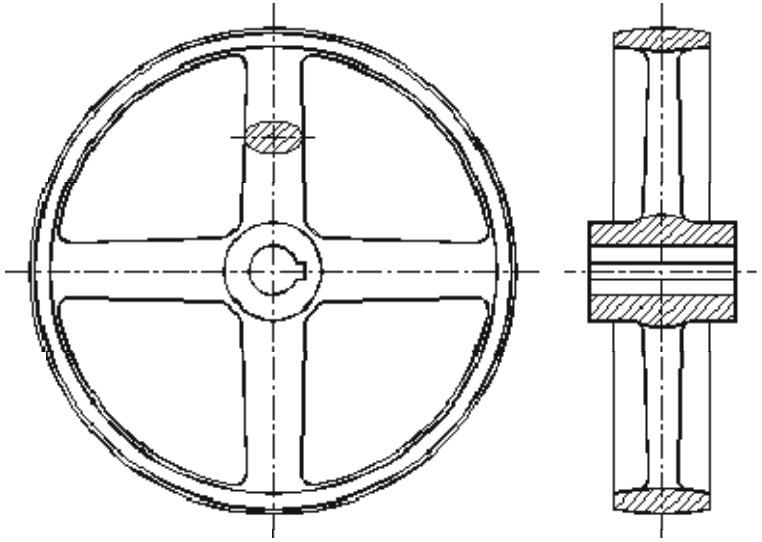
ELEMENTOS DE MÁQUINAS

Transmissão por correntes

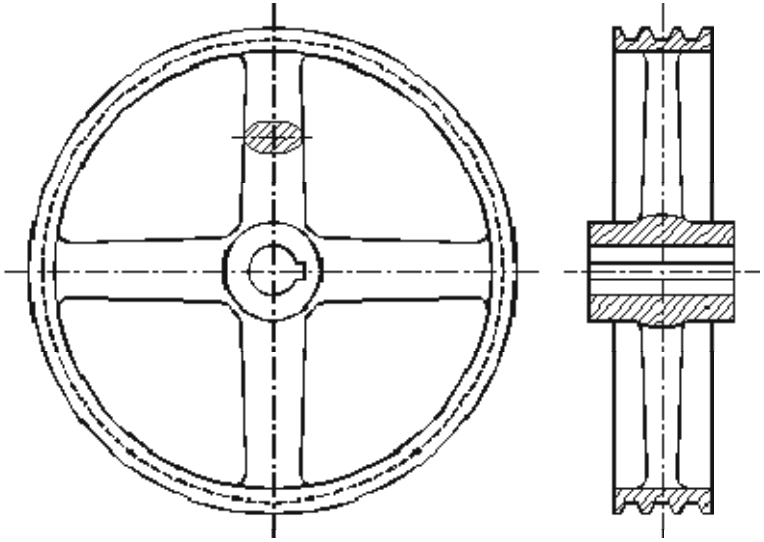


Transmissão por correias dentadas





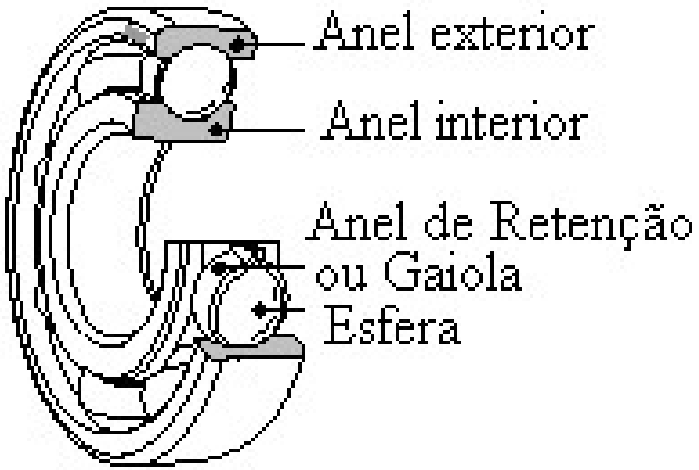
CORREIAS RECTANGULARES

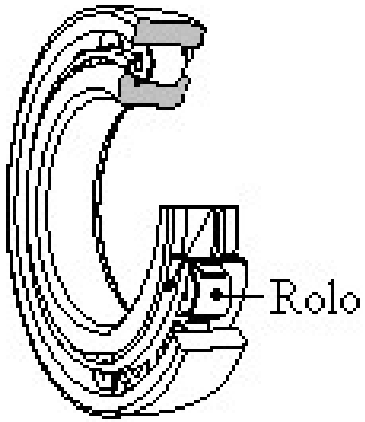


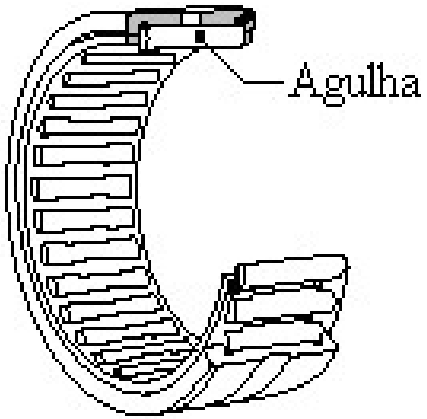
CORREIAS TRAPEZOIDAIS

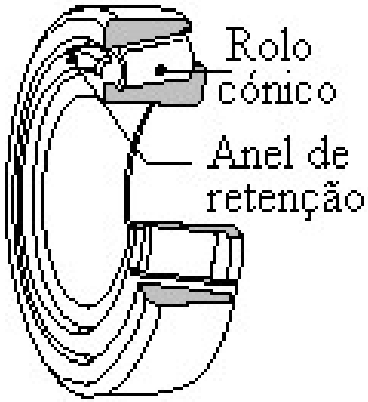
ELEMENTOS DE MÁQUINAS

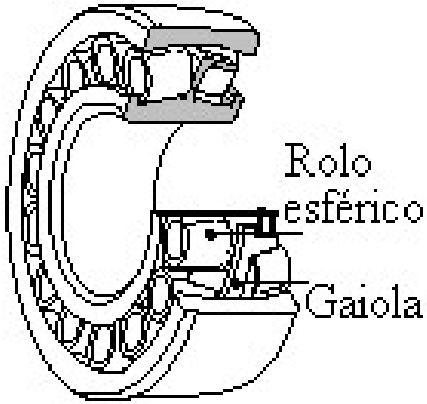
Rolamentos





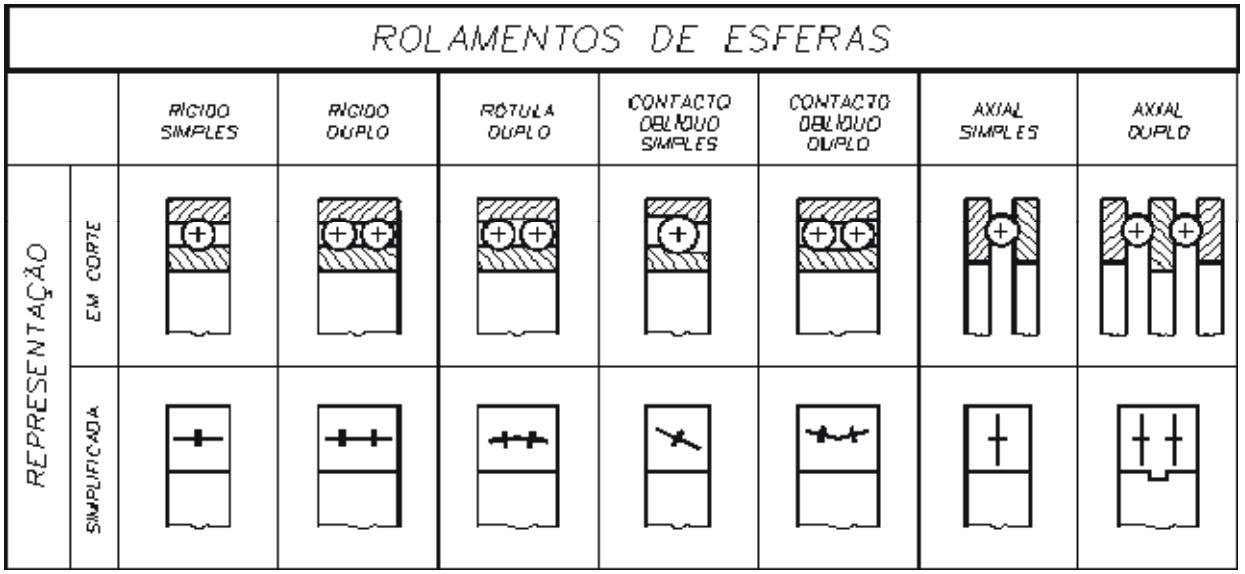


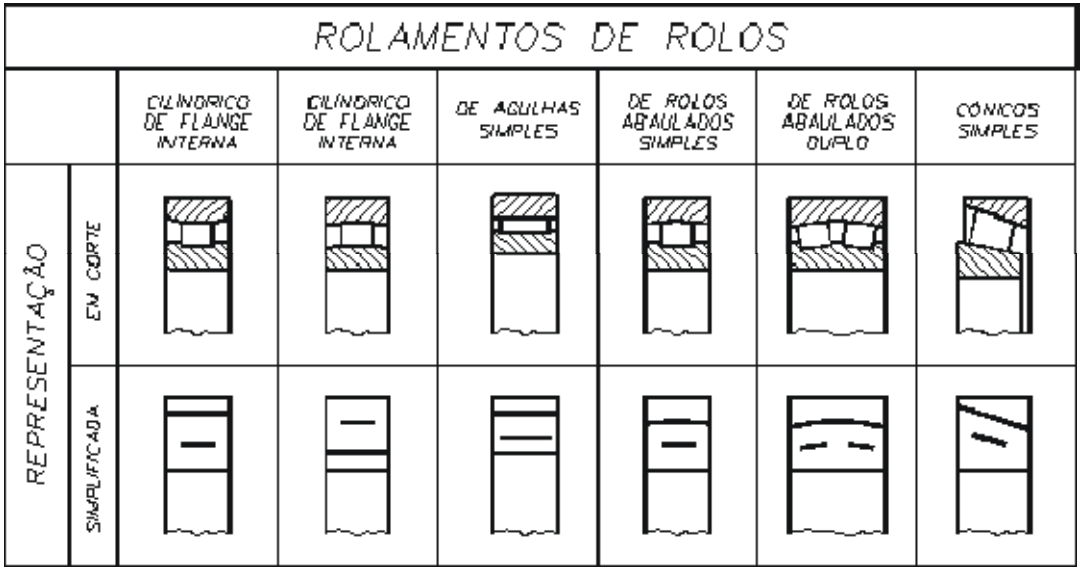




ELEMENTOS DE MÁQUINAS

Representação de Rolamentos





ELEMENTOS DE MÁQUINAS

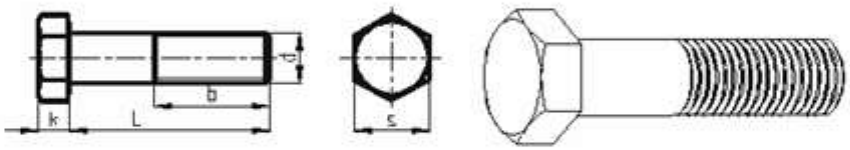
Especificação de componentes (ver Anexo B)

Parafuso Cabeça Hexagonal ISO4014 - M18x80 8.8

Desenho Técnico Moderno

B.1 PARAFUSOS

Designação:	Parafuso Cabeça Hexagonal		
Normas:	ISO 4014	DIN931	Classe 8.8



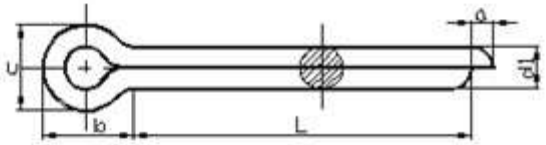
d	4	5	6	7	8	10	12	14	16	18	20	22	24	27	30	33	36	39	42	45	48	52	56	60	64
P	0.7	0.8	1	1	1.3	1.5	1.8	2	2	2.5	2.5	2.5	3	3	3.5	3.5	4	4	4.5	4.5	5	5	5.5	5.5	6
b	14	16	18	20	22	26	30	34	38	42	46	50	54	60	66	72	78	84	90	96	102	116	124	132	140
k	2.8	3.5	4	4.8	5.3	6.4	7.5	8.8	10	12	13	14	15	17	18.7	21	22.5	25	26	28	30	33	35	38	40
s	7	8	10	11	13	16	18	21	24	27	30	32	36	41	46	50	55	60	65	70	75	80	85	90	95
Lmin	25	30	30	30	35	40	45	50	55	60	65	70	75	80	90	100	110	110	140	120	140	160	160	160	180
Lmax	50	80	120	80	200	220	300	260	320	300	400	300	440	400	480	360	500	400	500	400	500	400	440	400	400
Linc	5	5	5	5	5	5	5	5	5	5	5	10	10	10	10	10	10	10	10	20	10	20	20	20	20

ELEMENTOS DE MÁQUINAS

Freio de Imobilização ISO1234 - 4x20

B.3 FREIOS

Designação:	Freios de Imobilização–dxL		
Normas:	ISO 1234	DIN 94	Em Aço



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d(max.)	0,9	1,4	1,8	2,3	2,9	3,2	3,7	4,2	4,6	5,9	6,5	7,5	9,5	12,4
d1(min.)	0,8	1,3	1,7	2,1	2,7	3	3,5	4	4,4	5,7	6,7	7,3	9,3	12,1
a(max.)	1,6	2,5	2,5	2,5	3,2	3,5	4	4	4	4	4,7	4	6,3	6,3
b	3	3,2	4	5	6,4	7	8	8,5	10	12,6	12,3	16	20	25
c(min.)	1,6	2,4	3,2	4	5,1	6,1	6,5	7,6	8	10,3	12,45	13,1	16,6	21,7
c(max.)	1,8	2,8	3,6	4,6	5,8	6,8	7,4	8,2	9,2	11,8	12,5	15	19	24,8
Lmin	10	10	16	16	16	20	20	50	25	25	63	40	63	63
Lmax	50	50	63	63	63	80	80	63	100	100	80	125	160	160
Linc	2-5	2-5	4-5	4-5	4-5	5	5	-						